

SerenDoc: A Human-Centric AI and IoT-Based Clinical Workload Orchestration System for Smart Healthcare

A PROJECT REPORT

Submitted by

PRAGADEESWARAN K [RA2211047010135]

SHYAM J [RA2212701010002]

ADITI V [RA2212701010020]

Under the Guidance of

Dr. Vinston Raja R

Assistant Professor, Department of Computational Intelligence

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of*

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COLLEGE OF ENGINEERING AND TECHNOLOGY
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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Degree/ Course : B. Tech – Artificial Intelligence
Student Name : PRAGADEESWARAN K, SHYAM J, ADITI V
Registration Number : RA2211047010135, RA2212701010002, RA2212701010020
Title of Work : SerenDoc: A Human-Centric AI and IoT-Based Clinical Workload Orchestration System for Smart Healthcare

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SIGNATURE

Dr. Vinston Raja R
Assistant Professor
Department Of
Computational Intelligence

Examiner 1

SIGNATURE

Dr. R. ANNIE UTHRA
Professor & Head
Department Of
Computational Intelligence

Examiner 2

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ABSTRACT

SerenDoc proposes a role-based, web-first hospital workflow system designed to reduce information friction across patient intake, monitoring, clinical documentation, and governance. In many care settings, operational delays and clinical risk arise not only from a lack of digitization, but from fragmented tools, inconsistent access control, and interfaces that do not match the day-to-day responsibilities of reception staff, nurses, doctors, interns, and administrators. Conventional approaches often rely on manual handoffs, repeated data entry, and delayed visibility of changes such as vital-sign deterioration, while auditability and privacy requirements demand stronger guarantees than client-side permission checks alone. Fundamentally, SerenDoc implements a unified patient–visit–document model with policy enforcement at both the user interface and the database. The system is delivered as a single-page application built with React and TypeScript, backed by Supabase services (PostgreSQL, Authentication, Storage, Realtime, and Edge Functions). Intake workflows support guided patient registration and visit creation, including document upload and OCR-assisted extraction for faster data capture. A Human-in-the-Loop verification step ensures that OCR outputs are reviewed and approved by staff before becoming authoritative records, supporting accuracy and accountability in sensitive clinical contexts. Beyond onboarding, SerenDoc extends to clinical and operational capabilities required for multi-role coordination: QR-based patient identification for quick context resolution, append-only nursing and clinician entries to preserve history, prescriptions and case sheets for structured documentation, and vitals monitoring with realtime updates and threshold-oriented alerts. Governance features include severity-aware access constraints (including intern restrictions for high-severity cases), case allocation aligned with role and policy, and tamper-evident audit and activity views that make key actions traceable without mixing routine operations with forensic logging. Administrative dashboards and analytics surfaces provide oversight, and an IoT fleet module supports device pairing and connectivity visibility where bedside monitoring hardware is deployed. By combining role-shaped user experiences, realtime observability, and server-enforced access control with reviewable automation (OCR), SerenDoc offers an integrated approach to hospital workflow digitization that emphasizes privacy, traceability, and usability under interruption-driven environments. The project demonstrates a practical, modular architecture that is deployable as a lightweight web application while remaining extensible for future enhancements such as deeper analytics, interoperability, and expanded device integration.

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ABBREVIATIONS

ADT	Admission, Discharge, Transfer
API	Application Programming Interface
CI	Continuous Integration
DBA	Database Administrator
DOB	Date of Birth
HTTPS	Hypertext Transfer Protocol Secure
ID	Identification / Identifier
IoT	Internet of Things
IPD	In-Patient Department
IT	Information Technology
KPI	Key Performance Indicator
MAR	Medication Administration Record
MRN	Medical Record Number
OCR	Optical Character Recognition
OPD	Out-Patient Department
PDF	Portable Document Format

PHI	Protected Health Information
QA	Quality Assurance
QR	Quick Response
RBAC	Role-Based Access Control
REST	Representational State Transfer
RLS	Row Level Security
RPC	Remote Procedure Call
Rx	Medical Prescription
SDG	Sustainable Development Goal
SOAP	Subjective, Objective, Assessment, and Plan (Medical Notes)
SpO2	Peripheral Capillary Oxygen Saturation
UI	User Interface
UN	United Nations
US	User Story

CHAPTER 1

INTRODUCTION

1.1 Introduction

Modern healthcare facilities operate as coordinated socio-technical systems in which outcomes depend as much on the flow of information as on clinical knowledge alone. Patients move through registration, triage or intake, nursing assessment, physician or trainee review, diagnostics, prescribing, and discharge or transfer. At each step, new data are generated: identifiers, chief complaints, vital signs, allocation decisions, medication orders, and narrative or structured notes. When these artifacts remain synchronised across roles and over time, teams can maintain continuity of care. When they are scattered across incompatible channels—paper binders, standalone spreadsheets, messaging threads without linkage to the record, or departmental databases that do not share a common patient key—the same clinical staff must spend effort reconstructing context rather than acting on it. That reconstruction is slow, error-prone under pressure, and difficult to audit afterward.

The problem is not solely one of “going digital.” Many institutions already use computers for billing, scheduling, or partial documentation. Difficulty often arises from a mismatch between how work is divided among roles and how software surfaces tasks. Ward and clinic work is interrupt-driven: a nurse may be pulled between bedside checks, documentation, and responding to abnormal values; a receptionist may need to complete intake while a queue forms; a physician may have minutes to prepare before entering a room. Tools that require long navigation paths, that show every function to every user, or that refresh only on manual reload are poorly aligned with that tempo. Likewise, regulatory and institutional expectations around privacy and accountability mean that access cannot be an afterthought: who may read or write which rows of data must be enforceable in a durable way, not only hidden behind menu items in the user interface.

Against that background, this report examines SerenDoc, a web-based application intended for use in ward and clinic environments where multiple professional roles collaborate around the same patients and visits. SerenDoc is organized around a single logical model of patients, visits, allocations, vitals, prescriptions, medical records, and case-sheet style documentation, while presenting role-appropriate screens so that each category of user sees work that corresponds to their responsibilities. Reception-oriented flows support registration and visit intake; nursing-

oriented flows support monitoring and quick identification, including QR-based resolution of patient context where the deployment enables it; clinical flows support review of longitudinal profiles, prescriptions, and structured case sheets; administrative flows support oversight, analytics where implemented, and visibility into connected devices when an IoT-oriented module is in use. Operational transparency is supported through activity logging and audit-oriented records, so that important actions can be reviewed after the fact without relying solely on individual memory.

Technically, the implementation described in the repository follows a client-heavy architecture. The user interface is built with Vite, React, and TypeScript, which supports component reuse, static typing for safer refactoring, and a development model familiar to contemporary web teams. Persistent data, authentication, row-level authorisation, realtime subscriptions, and selected server-side operations are provided through Supabase: a managed stack combining PostgreSQL with Row Level Security, authenticated client sessions, realtime channels suitable for vitals or alert scenarios, and Edge Functions for tasks that should not run entirely in the browser—such as OCR-assisted extraction from uploaded images or PDFs during registration, where staff remain responsible for verifying captured fields. That split is common for single-page apps that still enforce policy on the server for sensitive domains such as healthcare. Sections 1.2–1.7 take up motivation, links to selected UN Sustainable Development Goals, stakeholders, vision, product goal, and the Microsoft Planner-style backlog. SerenDoc is treated as support for coordination and documentation: it does not replace clinical judgment or local policy, but it can reduce friction around information.

1.2 Motivation

Care coordination failures in hospitals and clinics frequently present as delays, duplication, or ambiguity rather than as a single dramatic software bug. A vital sign may have been taken but not yet visible to the clinician preparing for review; an allocation may have changed while the printed list at a desk was still authoritative; a prescription update may live in one module while the nursing worklist still reflects an older state. In each case, individual actors may behave correctly within their own fragment of the process; the system as a whole still fails because there is no single, timely, authoritative view for authorized participants. Informal mitigation—telephone calls, verbal handoffs at shift change, or ad hoc photographs of monitor screens—

can preserve short-term continuity but does not produce structured, queryable evidence for quality improvement or for investigations when something goes wrong.

Role heterogeneity compounds the issue. Registration staff, nurses, physicians, trainees, and administrators do not share identical goals at every moment. The receptionist's priority is often completeness and speed of demographic and visit data capture; the nurse's priority is often detection of instability and timely escalation; the physician's priority is often synthesis of history, findings, and orders within limited time; the administrator's priority may include throughput, resource use, and compliance with internal protocols. A single undifferentiated dashboard tends to optimize for none of these: it either overloads everyone with the same widgets or pushes each role into deep menus to find what matters. SerenDoc's design motivation is therefore not only "digital records" but role-shaped surfaces backed by one consistent underlying representation of patients, visits, vitals, and allocations, so that handoffs become transitions between views on shared data rather than re-keying or re-fetching from parallel silos.

Security and traceability form a second pillar of motivation. Healthcare data are sensitive; unauthorized disclosure or improper modification can harm patients and expose institutions to legal and reputational risk. Technical expectations have moved beyond perimeter firewalls alone toward fine-grained access control at the data layer, short-lived credentials where appropriate, and explicit logging of reads and writes that matter for governance. Row Level Security in PostgreSQL, applied consistently with the application's role model, offers a way to enforce policies close to the data so that a misconfigured client route does not automatically imply a full breach of confidentiality. Complementary activity and audit logs address a different question: not only whether access was permitted, but what sequence of actions occurred when reconstructing an episode of care or responding to an internal review. A third motivation is temporal: monitoring use cases benefit when updates propagate without requiring users to poll or refresh. Vital signs and related alerts are classic examples where latency measured in tens of seconds can change triage behavior. Realtime channels, when used with appropriate authorisation, narrow the gap between "the value changed in the database" and "an authorized observer can see the change." That capability must be paired with clarity about who is subscribed, under what conditions, and how alert fatigue is managed in policy and UI design; the software supplies mechanisms, while institutions supply rules.

Finally, there is a pragmatic motivation rooted in cost and maintainability. Large integrated vendor suites can offer breadth, but they may be expensive to license, slow to customize, and dependent on vendor roadmaps. At the other extreme, purely bespoke stacks can burden small IT teams with operations they are not staffed to run. A browser-based client on widely understood frameworks, backed by a managed platform with clear boundaries between publishable client keys and privileged service roles, occupies a middle ground: it allows a student or institutional project to demonstrate a credible architecture without pretending to replace enterprise procurement decisions. Optional extensions—IoT fleet views, OCR at registration—illustrate how the same core entities can accrete new behaviors without rewriting the fundamental patient-visit model.

1.3 Sustainable Development Goal of the Project

United Nations Sustainable Development Goal 3 calls for ensuring healthy lives and promoting well-being for all at all ages. The goal is broad; for digital health projects, the most salient connections are usually found in targets that concern health system performance, access to quality services, and risk reduction rather than in any single line of code. Well-functioning information systems can support timely care, reduce redundant or conflicting orders when paired with good clinical process, and make follow-up more reliable by keeping visit and medication context visible across encounters. They can also support workforce conditions indirectly: if less time is spent hunting for results or re-entering demographics, more time may be available for direct patient contact, though such outcomes are contingent on implementation quality, training, and leadership—not guaranteed by deployment alone.

Relating SerenDoc specifically to SDG 3 is therefore a matter of plausible contribution rather than of claiming a measurable impact on population health statistics from a single academic software artifact. The system’s consolidation of registration, visits, vitals, documentation, and prescriptions into one logical model addresses a narrow but real part of the care pathway: the integrity and availability of operational and clinical-administrative data inside a facility. Where local processes are sound, reducing information gaps may support fewer conflicting instructions, faster recognition of changes in monitored variables, and clearer accountability during reviews. Where local processes are weak, software may change little until governance catches up; acknowledging that limit keeps the SDG discussion honest.

Sustainable Development Goal 9 concerns industry, innovation, and infrastructure—including resilient infrastructure and inclusive access to technological capacity. Healthcare institutions depend on digital infrastructure that is available, authenticated, and resilient to common failure modes such as credential leakage or accidental broad exposure of data. Architectures that use encrypted transport, authenticated APIs, database-enforced policies, and server-side functions for sensitive operations exemplify how “infrastructure” is implemented in practice. Incremental adoption paths—adding realtime monitoring or OCR without replacing every legacy workflow on day one—mirror SDG 9’s emphasis on innovation that can be integrated rather than only on greenfield megaprojects.

Equity sits at the intersection of SDG 3 and SDG-related debates about who benefits from innovation. Browser-based clients and managed backends can lower the barrier to running a consistent application across devices already owned by a facility, compared with thick clients that require specific workstations or lengthy installation cycles. They do not remove structural barriers such as unreliable connectivity, lack of training budgets, or understaffing. Digitalization can even worsen inequity if better-resourced sites capture disproportionate benefits. The project’s technical choices should therefore be read as enabling conditions for broader access to a certain class of tool, not as a complete equity strategy. This SDG discussion stays interpretive. SerenDoc supports coherent records, role-appropriate access, traceability, and timely updates in line with how many facilities describe better information systems. Measuring impact on SDG indicators would need empirical study, partner sites, and ethics review outside this report.

1.4 Product Vision Statement

1.4.1. Audience

1.4.1.1 Primary Audience:

- SerenDoc targets healthcare providers and facilities that coordinate intake, monitoring, and documentation across several roles in wards, clinics, or similar outpatient settings. These organizations range from smaller clinics with limited IT capacity to larger multi-department facilities that still benefit from a focused web application rather than a single undifferentiated enterprise screen. SerenDoc is intended to support continuity of information, timely visibility of changes (including vitals), and traceable actions, while integrating with a managed backend rather than requiring a bespoke server team for

every deployment. Key users include reception and front-office staff responsible for registration and visit routing; nursing staff responsible for vitals and monitoring-oriented worklists; physicians and interns responsible for review, case sheets, and prescribing; and administrators responsible for oversight, analytics, and—where used—connected device inventory.

1.4.1.2 Secondary Audience:

- The secondary audience includes hospital or clinic information technology and information security staff who evaluate single-page application architectures, session handling, Content Security Policy, separation of publishable keys from privileged service roles, and the use of Edge Functions for server-side processing. Educators and students in health informatics, biomedical engineering, or software engineering may use SerenDoc as a case study in role-based access enforced at both route and database (Row Level Security) layers, realtime clinical dashboards, and the distinction between compliance-oriented audit logging and user-facing activity timelines. Researchers interested in workflow evaluation, usability, or workload allocation under ethical approval may also draw on the system’s documented behavior and repository structure, without treating the academic artifact as a substitute for regulated clinical trials or population-health studies.

1.4.2. Needs

1.4.2.1. Primary Needs:

- **Coordinated patient and visit management:** One clear path from registration through visit creation, allocation, and status, so teams do not rely on informal lists that drift from the record.
- **Realtime monitoring and alerting:** Vitals and related signals should reach authorised staff quickly (realtime channels where configured), with alerts that support response instead of constant manual refresh.
- **Role-appropriate access and interfaces:** Each role sees work that fits the job—intake and routing for reception, monitoring and identification (including QR where enabled) for nursing, longitudinal views for doctors and interns, oversight for administration—with the same rules in the router and in database policies.

- **Traceability and operational transparency:** Logs of important actions and user-visible activity timelines so handoffs and reviews can reconstruct what happened without relying only on memory.

1.4.2.2. Secondary Needs:

- **Administrative analytics and throughput visibility:** Where implemented, dashboards show load, usage, and workflow health for operations leads rather than bedside triage.
- **Device and connectivity oversight:** IoT or similar views show paired devices when hardware is in use, so connectivity is not invisible to operations
- **Reduced duplicate data entry at intake:** OCR with mandatory staff review cuts retyping from scans and PDFs while keeping people responsible for final values.
- **Engineering assurance:** Tests, static typing, and checked environment config limit regressions as the codebase grows.

1.4.3. Products

1.4.3.1. Core Product:

- SerenDoc is a web-based, role-aware care workflow system implemented as a single-page application (Vite, React, TypeScript) backed by Supabase (authentication, PostgreSQL with Row Level Security, realtime subscriptions, and Edge Functions). The core product integrates patient registration and multi-step intake where configured, patient listing and profiles, visit intake, case allocation (including workload-oriented assignment patterns supported in the application), vitals monitoring with realtime updates, medical records, electronic prescribing flows, structured case sheets (including controlled public sharing where that feature is enabled), and operational activity logging. Access control is applied at the route level for navigation and at the database level through RLS aligned to institutional roles. The product is designed to be deployable as a responsive web client with a managed backend, reducing the need for thick-client installations on dedicated workstations while still keeping sensitive processing and policy enforcement on server-side paths where appropriate.

1.4.3.2. Additional Features:

- **OCR-assisted intake:** Server-side document processing via Supabase Edge Functions supports extraction from uploaded images or PDFs during registration, subject to validation by staff before data are committed as authoritative.

- **QR-based identification:** Nursing-oriented flows can resolve patient context quickly by scanning QR codes where the deployment uses that pattern, reducing lookup time at the point of care.
- **IoT device fleet views:** Administrative or clinical operations views can surface status of connected devices when an IoT fleet module is part of the environment.
- **Admin analytics:** Summaries and charts support institutional oversight of system usage and workflow indicators where those views are present in the implementation.
- **Security-oriented engineering practices:** Examples include Content Security Policy configuration in development, strict handling of environment variables at startup, and documented separation of publishable Supabase keys from service-role credentials used only in trusted scripts or server contexts.

1.4.4. Values

1.4.4.1. Core Values:

- **Clarity and tempo:** Interfaces favor glanceable status, short primary paths for frequent tasks, and limited purposeful use of trends over dense charting, consistent with high-interruption ward and clinic work.
- **Role integrity:** Permissions and visible actions reflect clinical and operational boundaries so that users are not exposed to irrelevant or prohibited functions solely because they share the same application binary.
- **Traceability:** Material actions leave discoverable records suitable for audit review and user-facing timelines, supporting accountability without treating logging as a substitute for clinical judgment.
- **Honest scope:** The system is positioned as coordination and documentation infrastructure; it does not claim autonomous diagnostic authority or regulatory certification by default.

1.4.4.2. Differentiators:

- **Dual-layer access control:** Combining React Router-level protection with PostgreSQL Row Level Security reduces the risk that a client-only check becomes the sole gate for sensitive reads and writes.
- **Realtime-first monitoring posture:** Managed realtime channels for vitals-oriented views avoid relying only on manual refresh like a plain request-response viewer.

- **Extensibility without fragmenting the record:** Edge Functions, modular feature folders, and optional IoT or OCR capabilities attach to the same patient–visit core rather than forcing parallel siloed mini-systems.
- **Verifiable engineering baseline:** TypeScript, automated tests, and explicit configuration validation support reproducible builds and clearer reasoning about behavior during academic evaluation or future maintenance.

1.5 Product Goal

The product goal for SerenDoc is a role-aware care workflow application that cuts unnecessary steps across registration, visits, monitoring, documentation, and prescribing while enforcing access rules and traceability. The repository and functional docs describe four strands: operations, clinical information, governance, and engineering. Operationally, the aim is fewer handoffs from arrival through registration, visit recording, allocation, and into monitored or documented care where those modules exist. Judging that fairly would need timed tasks or usability sessions against a baseline; this report does not include that measurement.

Clinically, authorised staff should share an up-to-date picture of the stay: identity, reason for visit, progress, vitals, prescriptions, and case-style documentation. Realtime updates shorten the gap between a change in monitored state and what staff are allowed to see; they do not replace clinical judgment about what the change means.

Governance means matching technical controls to institutional roles, keeping useful logs, and running sensitive work (bulk provisioning, OCR, and similar) with proper credentials and server-side code—not only in the browser.

Engineering means maintainable code: TypeScript, modular layout, automated tests, and a hard line between public client config and privileged secrets so behaviour can be described and refactored without silent breakage.

Explicit non-objectives matter for academic honesty. SerenDoc is not positioned as a clinical decision support engine that diagnoses disease autonomously. It does not, by itself, certify regulatory compliance in any jurisdiction. It does not replace training, staffing, or physical infrastructure. Success criteria for the project as a whole should therefore mix technical

completeness (features implemented, tests passing, policies defined) with qualitative assessment (expert review of workflows, heuristic usability feedback) unless the student secures access to operational metrics under proper agreements.

Further work can add monitoring, analytics, and document handling on the same patient–visit core instead of new silos.

1.6 Product Backlog

Table 1.1 Product backlog with user stories

S.No	User story
1	As a staff member, I want sign-up, sign-in, session lifecycle, email verification, and forgot/reset password flows on the platform identity service so that access is authenticated, recoverable, and consistent with security policy.
2	As the institution, I want role assignment enforced at routes and aligned with row-level database policies so that each role’s UI and data access stay consistent.
3	As a receptionist, I want guided multi-step patient registration together with visit intake (chief complaint, visit type, patient linkage) and validation so that intake and visit context are complete before downstream workflows depend on them.
4	As a receptionist, I want to upload PDFs and images during onboarding with type and size checks so that documents land in secure object storage tied to the patient context.
5	As a receptionist, I want OCR extraction from uploaded documents into registration fields with confidence cues so that data entry is faster while I remain accountable for final values.
6	As a receptionist, I want duplicate-patient hints during registration so that I can resolve likely matches before creating conflicting identities.
7	As a receptionist, I want a final review step to confirm or edit OCR-prefilled fields so that only verified information is persisted as the authoritative record.
8	As the platform, I want a relational patient–visit–document model with append-safe clinical storage and row-level security so that data stays queryable, protected, and traceable.

9	As staff in each role, I want patient lists, profile access, and a My patients worklist governed by the same rules as screens and row-level data so that viewing versus editing stays consistent and I only see patients I am allowed to care for.
10	As a nurse, I want nursing observations in an append-only medical record stream (no edit or delete of past rows), and as receptionist, doctor, intern, or nurse I want QR scan identification with safe failure on bad codes so that bedside verification is fast without exposing unnecessary patient detail.
11	As a doctor or intern, I want progress notes, diagnoses, and treatment captured in an add-only way, plus prescription flows (drug, dose, route, duration) with visit and provider linkage, so that clinical documentation stays traceable and prescriptions respect access policy.
12	As clinical staff, I want case sheets assembled from visits, diagnoses, prescriptions, and notes with a print-friendly view, reuse of document capture where OCR applies, and an optional minimal public case sheet link with create, renew, and cancel controls so that sharing is deliberate and compliant.
13	As the institution, I want patient severity (Low, Medium, High) with interns blocked on High severity in session, navigation, and allocation, with clear access-denied messaging and badges on lists and headers so that policy is visible before opening a chart.
14	As a nurse or clinician, I want vitals entry with consistent units, threshold-based alerts with sensible pacing where needed, monitoring dashboards, and realtime feeds so that abnormal values surface quickly and the UI stays usable under rapid updates.
15	As the platform, I want a central severity classification service fed by vitals with a backup calculation path when that service is unavailable, plus severity-change notifications and clear provenance of results, so that patient severity stays consistent without exposing sensitive health data in logs.
16	As care coordination staff, I want case allocation that considers severity, specialization, workload, and intern versus doctor rules, with automatic assignment where enabled, manual override, and safe behaviour under concurrency, so that patients are matched to appropriate providers.
17	As the institution, I want a tamper-evident audit trail with who, when, what, and record identifiers, and a separate operational activity timeline for reception and

	administration, so that governance and day-to-day operations are both visible without mixing forensic audit with routine activity.
18	As staff and administrators, I want role dashboards, an admin home with analytics, IoT fleet views for authorised roles, and safe settings for signed-in users, so that each role lands on relevant metrics and device health without leaking unnecessary data.
19	As a doctor, intern, or administrator, I want controlled patient and visit status transitions that align with severity and write audit entries, so that workflow state stays coherent under concurrent updates.
20	As the delivery team, I want automated checks, security validation, accessibility on primary flows, and a documented happy path for demonstration, so that release quality and stakeholder demos are repeatable without sensitive data in artefacts.

Table 1.1 matches tasks tracked in Microsoft Planner (buckets such as Product Backlog, Sprint Backlog (In-Progress), Awaiting Review, Completed), with priority, dates, notes, checklists, and links to dev work for January–April 2026.

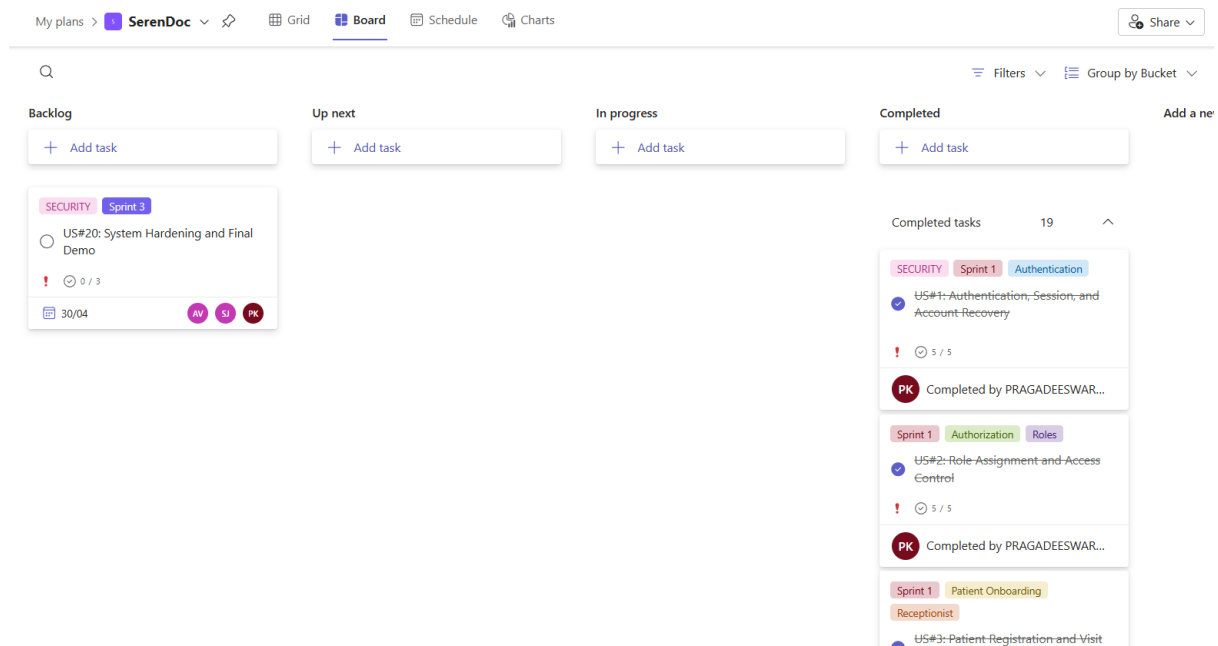


Figure 1.1 MS Planner Board of SerenDoc

1.7 Product Release Plan

The following Figure 1.2 depicts the release plan of the project

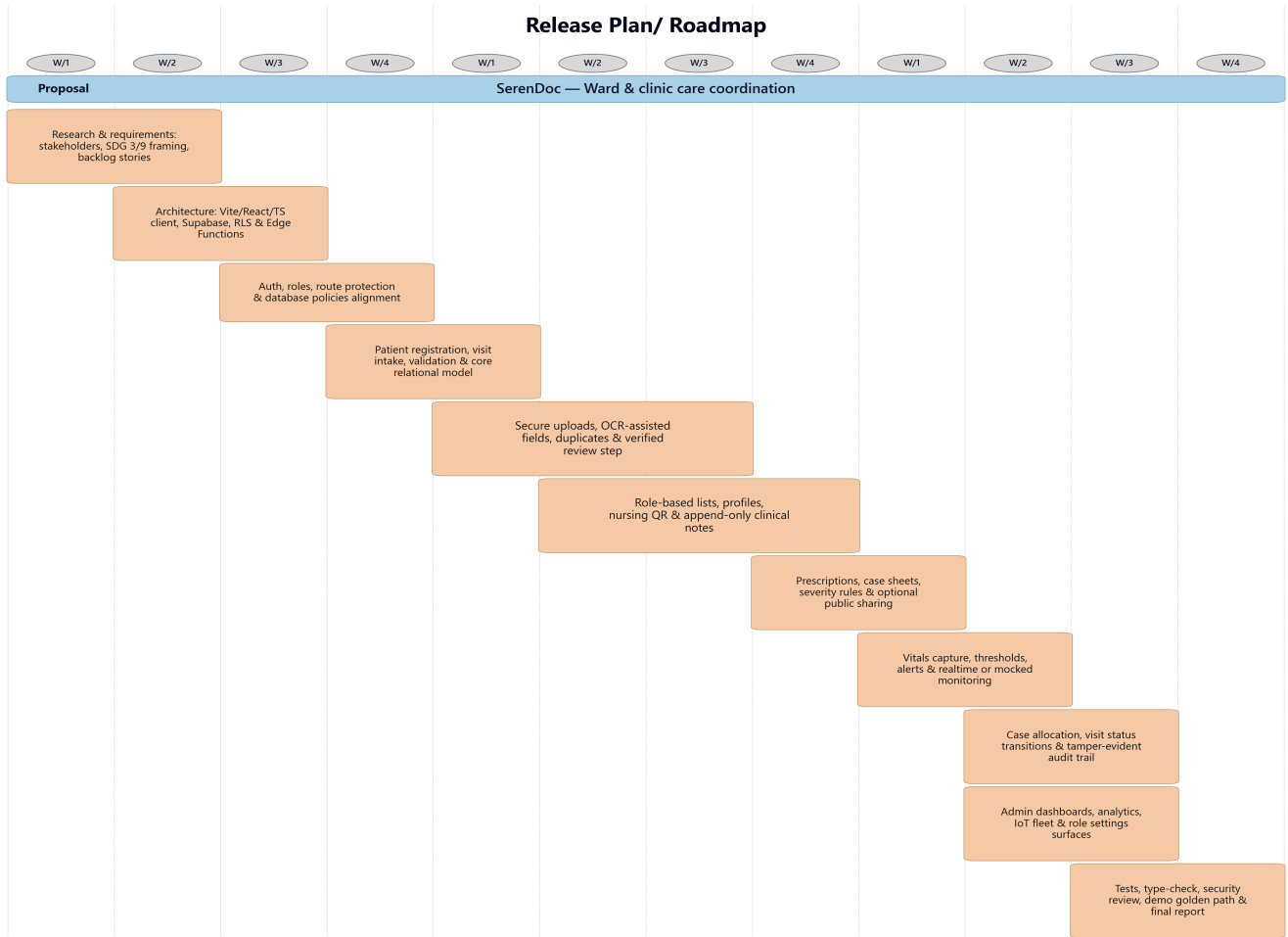


Figure 1.2 Release plan of SerenDoc

CHAPTER 2

SPRINT PLANNING AND EXECUTION

2.1 Sprint 1

2.1.1 Sprint Goal with User Stories of Sprint 1

To build the foundational layer of SerenDoc by enabling staff to sign in securely, work under role-based access, register patients (including visit intake), upload documents, run OCR with receptionist review and duplicate checks, and store data in a protected patient-visit document model. This sprint (Table 2.1) ensures that core onboarding, authentication, authorisation, and storage are operational and ready for later clinical features.

Table 2.1 Detailed User Stories of sprint 1.

S.No	User story
#US 1	As a staff member, I want sign-up, sign-in, session lifecycle, email verification, and forgot/reset password flows on the platform identity service so that access is authenticated, recoverable, and consistent with security policy.
#US 2	As the institution, I want role assignment enforced at routes and aligned with row-level database policies so that each role's UI and data access stay consistent.
#US 3	As a receptionist, I want guided multi-step patient registration together with visit intake (chief complaint, visit type, patient linkage) and validation so that intake and visit context are complete before downstream workflows depend on them.
#US 4	As a receptionist, I want to upload PDFs and images during onboarding with type and size checks so that documents land in secure object storage tied to the patient context.
#US 5	As a receptionist, I want OCR extraction from uploaded documents into registration fields with confidence cues so that data entry is faster while I remain accountable for final values.

S.No	User story
#US 6	As a receptionist, I want duplicate-patient hints during registration so that I can resolve likely matches before creating conflicting identities.
#US 7	As a receptionist, I want a final review step to confirm or edit OCR-prefilled fields so that only verified information persisted as the authoritative record.
#US 8	As the platform, I want a relational patient–visit–document model with append-safe clinical storage and row-level security so that data stays queryable, protected, and traceable.

The Planner Board representations of the user stories are shown in the figures below.

... ✕

SerenDoc

 **US#1: Authentication, Session, and Account Recovery**

Completed on 02/02/2026 by you

 SECURITY ✕ Sprint 1 ✕ Authentication ✕

Bucket

Completed ▾

Progress

 Completed ▾

Priority

 Important ▾

Start date

01/01/2026 

Due date

14/01/2026 

Repeat

 Does not repeat ▾

Notes ☐ Show on card

- Organization **identity and sign-in** using standard secure practices: password rules, session handling (including refresh and sign-out), and clear error messages
- Dedicated flows for email verification, forgot password, and reset password; fair use limits on reset requests; clear redirects and email content aligned with organizational templates
- Sensitive configuration stays on trusted systems only; nothing privileged is exposed in end-user devices or public materials

Checklist 4 / 5 ☒ Show on card

-  Document and deliver primary sign-in and sign-up experiences
-  Session behavior (signed-in state, renewal, sign-out) and understandable auth errors
-  Verify-email, forgot-password, and reset-password experiences with secure one-time links and...

Figure 2.1 User story: Authentication, Session and Account Recovery

...

×

SerenDoc

✓

US#2: Role Assignment and Access Control

Completed on 02/02/2026 by you

AV

PK

Sprint 1

×

Authorization

×

Roles

×

Bucket

Completed

▼

Progress

✓

Completed

▼

Priority

!

Important

▼

Start date

05/01/2026

Due date

15/01/2026

Repeat

↺

Does not repeat

▼

Notes

☐ Show on card

• Staffroles: **Receptionist, Nurse, Doctor, Intern, Administrator**; role stored on each staff profile; enforce access **by screen and feature** and **by row** (who may see which patient records) with the same rules everywhere

• Receptionist: registration and visit intake; Administrator: analytics, device fleet, and admin views

Checklist 5 / 5

☐ Show on card

✓

Role list consistent across the product and stored data

✓

Protected areas and allowed actions per role

✓

Data access rules aligned with the screens for patients, visits, and documents

✓

Verification of the permission matrix for all five roles

✓

Document role capabilities in the internal wiki or product overview

Figure 2.2 User story: Role Assignment and Access Control

SerenDoc

✓ US#3: Patient Registration and Visit Intake

Completed on 02/02/2026 by you



🏷 Sprint 1 ✕ Patient Onboarding ✕ Receptionist ✕

Bucket	Progress	Priority
Completed ▾	✓ Completed ▾	! Important ▾
Start date	Due date	Repeat
08/01/2026 📅	22/01/2026 📅	🔄 Does not repeat ▾

Notes

☐ Show on card

- **Multi-step** registration: demographics, medical history, optional document capture path, final review; required fields checked before save
- **Visit flow**: chief complaint, visit type, link to the correct patient; receptionist-only where policy requires; handoff to vitals and allocation

Checklist 4 / 4

☒ Show on card

- ✓ Field set matches clinical intake (including emergency contact)
- ✓ Accessible forms with clear validation and save/update behavior with helpful errors
- ✓ Visit types (first visit, follow-up, emergency) per policy; who may create vs. who may view
- ✓ Success and failure messages; checks by role

Figure 2.3 User story: Patient Registration and Visit Intake

...

×

SerenDoc

✓

US#4: Document Upload System

Completed on 02/02/2026 by you

+

SJ

PK

📁

Sprint 1

×

Document Management

×

Upload

×

Bucket

Completed

▼

Progress

✓ Completed

▼

Priority

! Important

▼

Start date

10/01/2026

📅

Due date

20/01/2026

📅

Repeat

↺ Does not repeat

▼

Notes

☐ Show on card

- PDF and images; file type and size limits on forms and on the service; **managed file storage** with clear patient/onboarding folders; progress and preview; no sensitive health information in casual logs

Checklist 3 / 3

✓

Drag-and-drop and file picker with keyboard access

✓

Validation on upload; storage layout, access rules, and secure upload path

✓

Progress and error states; retention and access rules

☐ Show on card

Figure 2.4 User story: Document Upload System

SerenDoc

✓ US#5: OCR Pipeline for Patient Documents

Completed on 02/02/2026 by you



📁 Sprint 1 ✕ OCR ✕ Data Extraction ✕

Bucket

Completed ▾

Progress

✓ Completed ▾

Priority

! Important ▾

Start date

12/01/2026 📅

Due date

25/01/2026 📅

Repeat

🔄 Does not repeat ▾

Notes

☐ Show on card

- Signed-in requests to **document text extraction**; map labels to registration fields; confidence scores and low-confidence highlighting; timeouts, partial results, and cancel without leaving bad partial data

Checklist 3 / 3

☒ Show on card

- ✓ Document the agreed inputs and outputs (including how the signed-in user is identified)
- ✓ Normalize values (for example dates) in the form; per-field confidence in the multi-step flow
- ✓ Operational logging without sensitive content; sample files without real patient data

Figure 2.5 User story: OCR Pipeline for Patient Documents

SerenDoc

US#6: Patient Duplicate Detection

Completed on 04/03/2026 by SHYAM J (RA2212701010002)



Sprint 1 Patient Matching Data Integrity

Bucket	Progress	Priority
Completed	Completed	Important
Start date	Due date	Repeat
15/01/2026	28/01/2026	Does not repeat

Notes ☐ Show on card

- Debounced hints on name, phone, email; staff choose proceed or adjust; optional future: merge workflow — document only if committed

Checklist 3 / 3 ☒ Show on card

- Match thresholds and DB routine parameters with DBA review
- Debounced UI; degradation when duplicate service unavailable
- Tests for typos/transliteration

Figure 2.6: User story: Patient Duplicate Detection

...

×

SerenDoc

✓

US#7: OCR Verification and Approval Flow

Completed on 04/03/2026 by SHYAM J (RA2212701010002)

AV

SJ

PK

Sprint 1

×

OCR

×

Verification

×

Bucket

Completed

▼

Progress

✓ Completed

▼

Priority

● Medium

▼

Start date

18/01/2026

Due date

31/01/2026

Repeat

↺ Does not repeat

▼

Notes

☐ Show on card

- Receptionist review aligned with final profile; low-confidence flagged; intentional submit only; abandoned flows do not create incomplete live patient records

Checklist 3 / 3

✓

Show on card

✓

Review layout matches post-registration profile; inline edits and re-checking

✓

Discard/back without corrupting drafts; full path: upload → extraction → edit → submit

✓

Receptionist sign-off

Figure 2.7 User story: OCR Verification and Approval Flow

...

×

SerenDoc

✓

US#8: Patient Data Model and Storage

Completed on 04/03/2026 by SHYAM J (RA2212701010002)

+

AV

PK

📁

Sprint 1

×

Database

×

Data model

×

Bucket

Completed

▼

Progress

✓ Completed

▼

Priority

! Important

▼

Start date

01/01/2026

📅

Due date

20/01/2026

📅

Repeat

🔄 Does not repeat

▼

Notes

☐ Show on card

- Structured **patients**, **visits**, document metadata, **medical records** (add-only history); extracted fields searchable where possible; non-destructive handling of clinical content

Checklist 3 / 3

✓

Show on card

✓

Core records and relationships; access rules per entity; planned upgrades to storage

✓

Fast-enough search for allocation and duplicate detection; backup and restore expectations

✓

Integrity checks by role

Figure 2.8 User story: Patient Data Model and Storage

2.1.2 Functional Document

2.1.2.1. Introduction

This document outlines the core functionality delivered in Sprint 1 of SerenDoc. The sprint establishes the foundation for hospital management: secure staff authentication, role-based access, patient registration (including visit intake), document upload, OCR-assisted data extraction with receptionist review and duplicate checks, and storage in a protected patient–visit–document model. Staff work through a web application backed by Supabase (PostgreSQL, Storage, Realtime, Edge Functions), with visit registration and activity logging on the same platform.

2.1.2.2. Product Goal

The primary goal of this sprint is to deliver a secure, efficient patient onboarding system that reduces manual entry while keeping data trustworthy. The platform aims to:

- Enable receptionists to register patients through structured forms and through document upload with OCR.
- Support PDF and image uploads with validation, secure storage, and progress feedback in the UI.
- Use OCR (e.g. Tesseract or Google Vision API) to extract demographics and identifiers, with a mandatory review step before data persisted.
- Detect and surface likely duplicate patients so staff can resolve matches before creating conflicting records.
- Enforce role-based access so each role's routes and data visibility stay aligned with policy.
- Provide a responsive web experience with real-time updates where appropriate (e.g. OCR status via Supabase Realtime).

2.1.2.3. Demography (Users, Location)

Users:

- Receptionists: primary users for registration, document upload, OCR verification, and duplicate resolution.
- Administrators: oversight, analytics, and device fleet views where implemented; not the primary onboarding role unless policy assigns otherwise.
- Doctors: read access to patient records created in this sprint (no edit in Sprint 1).
- Nurses and interns: read-only access to patient records for care coordination, within role limits.

Location:

- Hospitals and similar healthcare facilities (single or multi-site), with on-premises or cloud deployment depending on environment.

2.1.2.4. Business Processes

- Staff authenticate via email and password; roles are assigned at registration and enforced on routes and data.
- Receptionists register patients manually or by uploading documents; files are validated and stored in secure object storage tied to the patient context.
- OCR runs on uploaded documents; extracted fields are shown for review before commit.

- Duplicate checks run against existing records (e.g. name, phone, ID number, date of birth); likely duplicates are flagged for resolution.
- Approved data is written to the patient master profile and document repository using append-safe patterns.
- Visit registration and activity/audit logging use the same Supabase backend

2.1.2.5. Features

This project focuses on implementing the following key features:

Feature 1: Patient onboarding and document upload

1. Description

- Receptionists capture patient demographics and visit context through guided multi-step registration and visit intake.
- PDF and image uploads are supported with type and size checks, upload progress, and preview; files are stored securely and linked to the patient.

2. User story

- As a receptionist, I want to register patients and upload PDFs or images during onboarding so that documents are stored safely and tied to the correct patient before downstream workflows use them.

Feature 2: OCR extraction, verification, and duplicate handling

1. Description

- OCR extracts structured fields (e.g. name, demographics, identifiers) from uploaded documents.
- A review step lets the receptionist correct values and approves only verified data; fuzzy matching highlights possible duplicates for resolution before saving.

2. User story

- As a receptionist, I want OCR to pre-fill registration fields with confidence cues and duplicate hints, and I want a final review step so that only verified information becomes the authoritative record.

Feature 3: Secure web application with RBAC and real-time interaction

1. Description

- Staff use a React web client that talks to Supabase over HTTPS with Row Level Security (RLS).
- Middleware enforces roles (Receptionist, Nurse, Doctor, Intern, Administrator); authenticated users see appropriate screens and data.
- Long-running steps (e.g. OCR) can be driven via Edge Functions with status conveyed through Realtime or polling, so the UI remains usable.

2. User story

- As a staff member, I want to sign in securely and interact with the system through a web interface that respects my role and shows onboarding and document status without unnecessary page refreshes.

2.1.2.6. Authorisation Matrix

Table 2.2 Access Level Authorisation Matrix of sprint 1

User role	Register patients / upload documents / run OCR review	View patient records	Approve OCR & resolve duplicates
Receptionist	Yes	Yes	Yes
Administrator	No	Yes	No
Doctor	No	Yes	No
Nurse	No	Yes	No
Intern	No	Yes	No
Unauthenticated	No	No	No

2.1.2.7. Assumptions

- Staff use supported browsers and a stable network; Supabase project, Storage, and Edge Functions are available in the target environment.
- OCR depends on a configured engine (e.g. Tesseract or Google Vision API) and acceptable document quality; poor scans may require manual correction at review.
- Sprint 1 focuses on foundational onboarding and storage; advanced clinical workflows are deferred to later sprints.
- Requirements are agreed for the sprint window; major scope changes are handled outside this baseline.

- Visit registration and activity log are implemented on the same Supabase backend as onboarding, consistent with the Sprint 1 architecture.

2.1.3 Architecture Document

2.1.3.1 Application

SerenDoc in Sprint 1 follows a microservices-style decomposition implemented on Supabase as the unified backend: Authentication, user role management, patient onboarding, document management, OCR processing, patient matching, and data storage each have a clear responsibility. The React frontend uses the Supabase client to reach PostgreSQL APIs, Storage, Realtime, and Edge Functions—there is no separate custom API gateway in this sprint.

Key logical modules include:

- **Authentication service** — Sign-up, sign-in, sessions, and tokens.
- **User role management** — RBAC and route protection aligned with database policies.
- **Patient onboarding service** — Manual entry and document-based flows, validation, profile creation.
- **Document management service** — Uploads, validation, Storage integration, previews.
- **OCR processing service** — Extraction pipelines (e.g. Edge Function ocr-process), mapping into the patient model.
- **Patient matching service** — Duplicate detection and merge-oriented workflows.
- **Data storage service** — Patient profiles, document metadata, version-safe append behaviour, RLS.

Visit registration and activity/audit logging share this backend

2.1.3.2. System Architecture

Figure 2.9 is the system architecture diagram of sprint 1

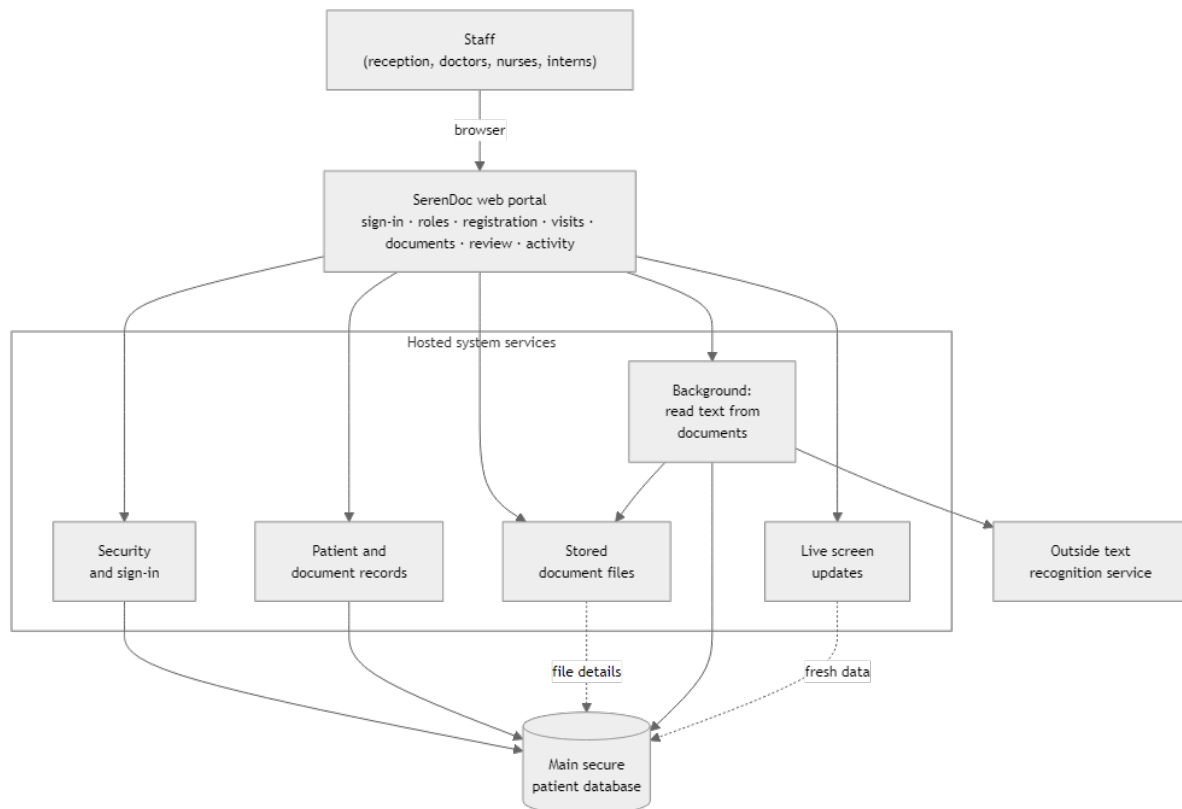


Figure 2.9: System Architecture Diagram (Sprint 1)

Summary:

Staff open SerenDoc in a web browser. The web portal connects to hosted services for sign-in, patient and document records, file storage, background reading of text from documents (using an outside recognition service when needed), and live updates on screen. A main secure patient database holds the authoritative records; file details and update feeds link documents and live views to that store.

2.1.3.3. Data Exchange Contract

Frequency of Data Exchanges:

Real-Time Exchanges:

- **Interactive (request–response):** Sign-in, registration, form saves, document upload initiation, and duplicate checks expect timely responses over HTTPS.
- **Asynchronous / background:** OCR and heavier matching may run in Edge Functions; the UI is updated via Realtime events, polling, or status fields rather than blocking the browser indefinitely.

- **Audit and activity:** Activity and audit data are recorded in line with application flows (e.g. via `log_audit_trail`), suitable for traceability in a clinical-adjacent setting.

Data Sets:

- **User data** — Email, name, role, credentials references, session tokens for authentication and authorisation.
- **Patient data** — Demographics, contact, visit linkage, medical history fields as defined in Sprint 1 schemas.
- **Document data** — Files (PDF/images), metadata, storage object references, processing status.
- **OCR extraction data** — Extracted fields, confidence, raw output; held until receptionist approval, then persisted.
- **Matching data** — Criteria and scores for duplicate detection and merge decisions.

Mode of Exchanges:

- **Supabase REST APIs** — Primary channel from React to Postgres-backed resources; RLS enforces access.
- **Supabase Storage API** — Multipart uploads to buckets; policies constrain who can read/write objects.
- **Supabase Edge Functions** — HTTP-triggered processing such as OCR; optional cron or webhooks as configured.
- **Supabase Realtime** — Subscriptions for status and notifications where the UI must update without full page reload.
- **PostgreSQL RPCs/transactions** — Used where atomicity and server-side logic are required (e.g. matching and audit logging).

2.1.4 UI DESIGN

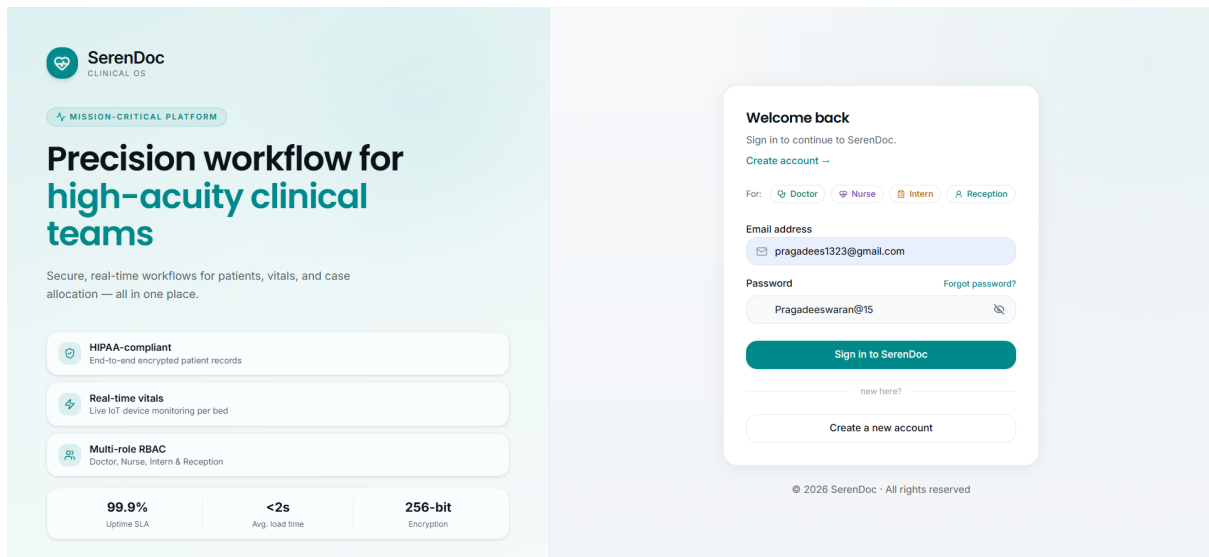


Figure 2.10 UI Design for Sign in Page

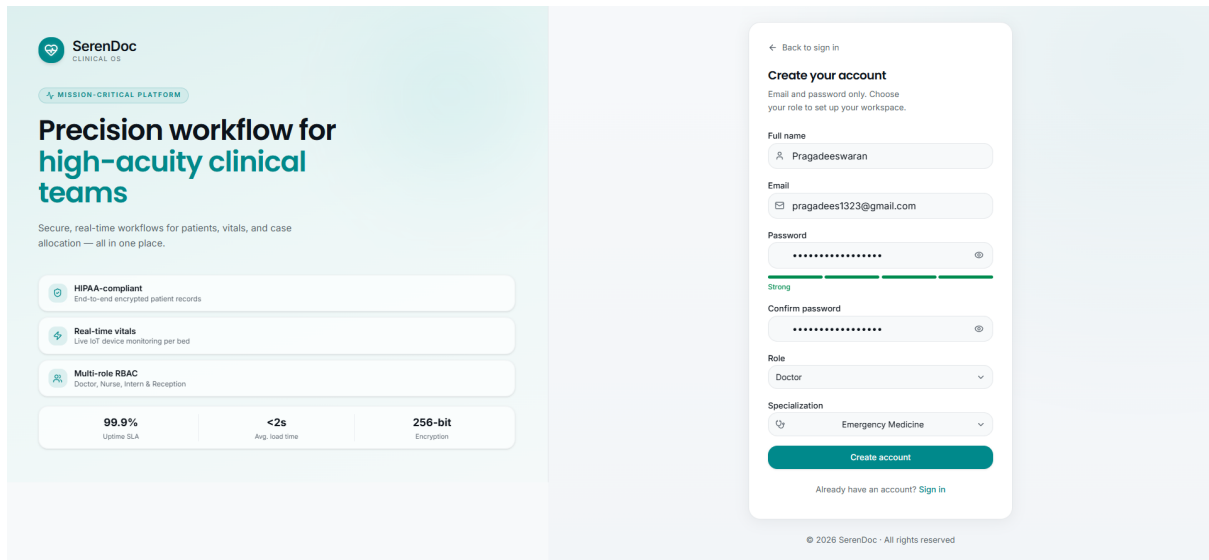


Figure 2.11 UI Design for Signup Page

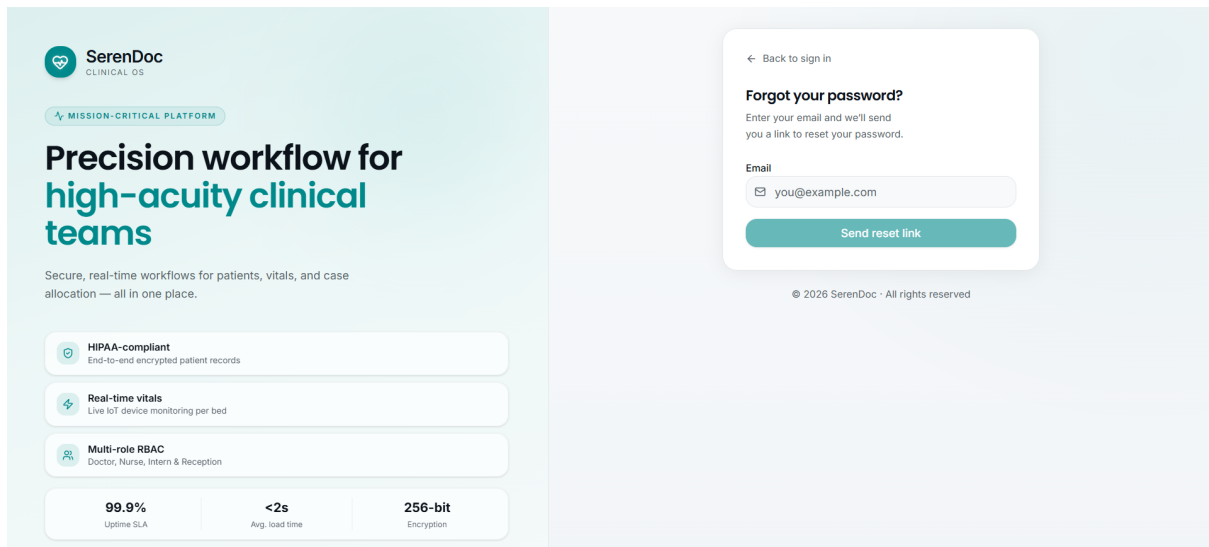


Figure 2.12 UI Design for Forgot Password Page

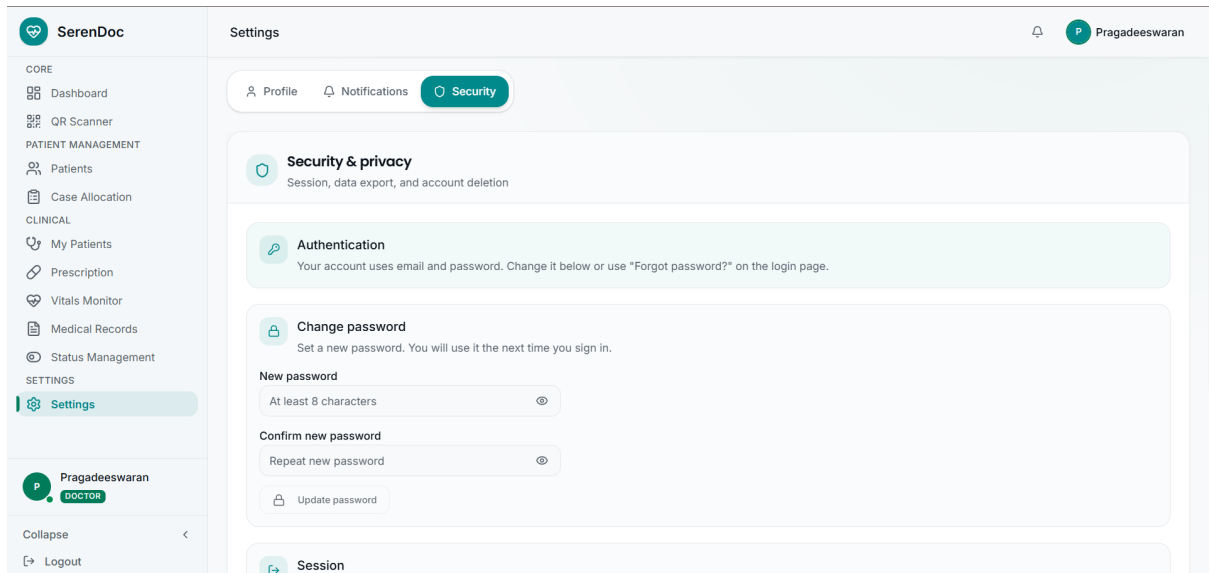


Figure 2.13 UI Design for Reset Password Page

The screenshot shows the 'New Patient Registration' page in a web application. The page has a sidebar on the left with icons for various functions. The main content area is titled 'New Patient Registration' and includes a sub-header 'Capture demographics, history, and review before creating the patient profile'. A progress bar at the top indicates 'Step 1 of 3' with a '33% complete' status. The steps are: 1. Patient Info, 2. Medical History, and 3. Review & Submit. The current step, 'Patient Info', features a 'Smart document scan' section. This section has a dashed box for uploading a document, with instructions: 'Drag & drop an ID card, prescription, or old record — or click to browse. We'll extract patient details automatically.' Below this are buttons for 'Upload document' and 'Enter manually'. A note at the bottom says 'Or fill the form out manually below.'

Figure 2.14 UI Design for Patient Registration (OCR Upload Step)

The screenshot shows the 'Patient Demographics' form in the 'Patient Registration' section. The form is titled 'Patient Demographics' and includes a sub-header 'Basic information about the patient'. The form is divided into several sections: 'Patient Type' with radio buttons for 'Out-Patient (OPD)' and 'In-Patient (IPD)'; 'Full Name' and 'Contact Number' fields; 'Date of Birth' and 'Age' fields; 'Gender' and 'Blood Group' dropdown menus; 'Contact & Personal Information' section with 'Email' and 'Address' fields; 'Occupation' and 'Annual Income' fields; and an 'Emergency Contact' section with 'Name', 'Relation', and 'Phone Number' fields. The form has a 'Back' button and a 'Next' button at the bottom.

Figure 2.15 UI Design for Patient Registration (Demographics step)

Patient Management >
Patient Registration

PR

Pragadees

Medical History

Current health concerns and medical background

Chief Complaint *

Patient suffers from constant diarrhoea

Main reason for this visit

History of Present Illness

Mental illness and Panic attacks

Detailed description of current symptoms and timeline

Pre-existing Medical Conditions

Select any conditions the patient currently has or has had in the past

☒ Diabetes
☒ Hypertension / High BP

☒ Asthma
☐ Cardiac Disorder

☐ Thyroid Disorder
☐ Kidney Disease

☐ Liver Disease
☐ Cancer

☐ Epilepsy
☐ Arthritis

Known Allergies

e.g., Penicillin, Peanuts, Dust

List any drug, food, or environmental allergies

Figure 2.16 UI Design for Patient Registration (Medical History step)

Patient Management >
Patient Registration

PR

Pragadees

Step 3 of 3

100% complete

☒ Patient Info

☒ Medical History

3

Review & Submit

Review Patient Information

Please verify all details before submitting

Demographics

Edit

Patient Type: Out-Patient (OPD)

Full Name: SHYAM.J

Age / Gender: 21 / Male

Date of Birth: 3/14/2005

Blood Group: O+ve

Contact Number: 1234567890

Email: shamcool@gmail.com

Address: ST, thomas mount road.

Occupation: N/A

Annual Income: N/A

Emergency Contact

Edit

Name:

Number:

Relationship:

Figure 2.17 UI Design for Patient Registration (Review & Submit step)

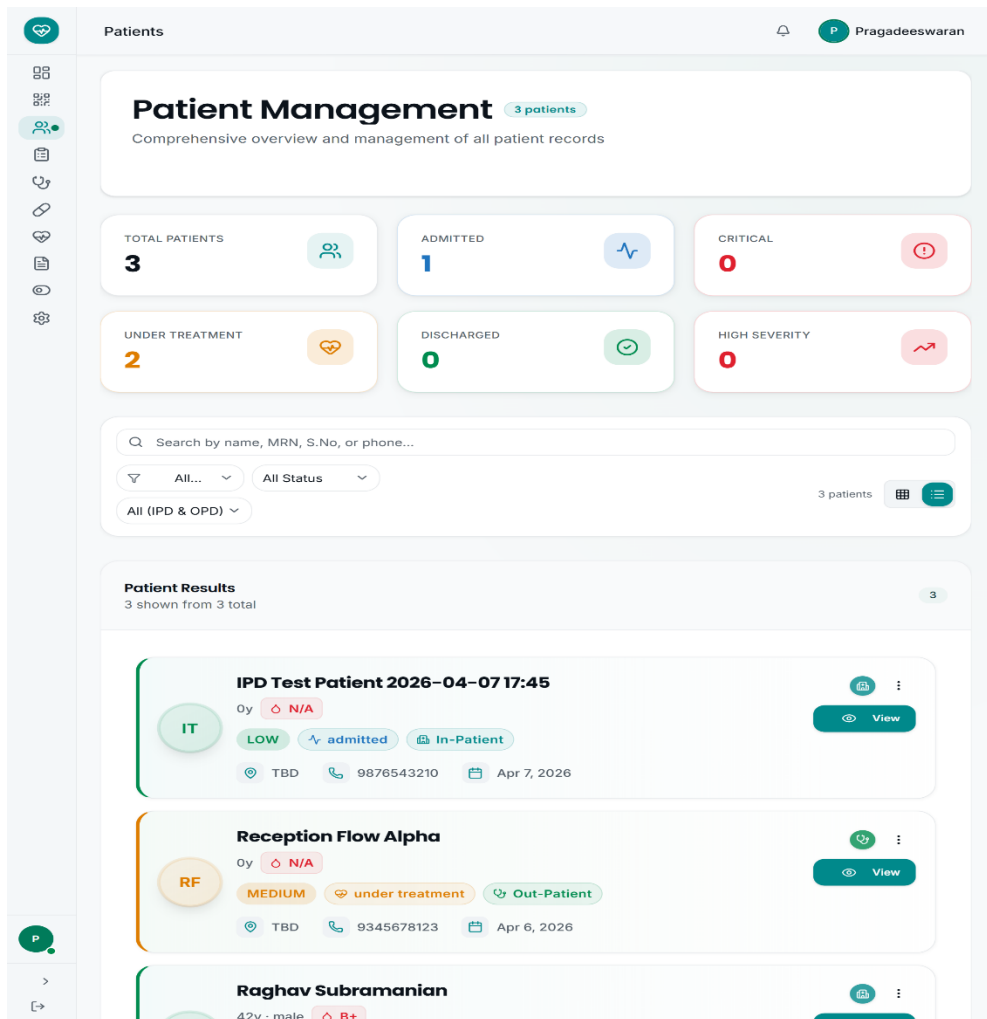


Figure 2.18 UI Design for Patient List Page (Patient Management)

Patient Management

Visit Intake

PR Pragadees

1 Find patient

2 Register visit

Patient Visit Registration

Register visit for Raghav Subramanian

Scan different patient

Standard flow: Find patient → Capture visit details → Route to nurse/doctor.

RS

Raghav Subramanian

In-patient

OP No 0P-456321 Age 42 yrs Gender Male Contact 9840123456 Email raghavs1983@gmail.com

Register visit

Case sheet

Visit details

Visit type *

First visit / New patient

Reason for visit (details) *

e.g. Follow-up for fever, routine check-up...

Briefly describe why the patient is here today.

Doctor assignment

Required specialization *

General Medicine

Preferred doctor (optional)

Pragadeeswaran — General Medicine

Start over

Register visit → Assign doctor

PR

>

↔

Figure 2.19 UI Design for Visit Intake Page

34

2.1.5 Functional Test Cases

Table 2.3 Detailed Functional Test Case of Sprint 1

Feature	Test Case	Steps to Execute Test Case	Expected Output	Actual Output	Status	More Information
Staff sign-in	Valid login	Open sign-in; enter email and password; submit.	Signed in; opens main dashboard.	Signed in; dashboard opened.	Passed	Session stays until sign-out.
Staff sign-in	New account and role	Register with name, email, password, job role; doctor adds specialty if asked; submit.	Account with role; can sign in.	Account OK; role OK; sign-in OK.	Passed	Doctors must give specialty when required.
Access by role	Who may open what	As nurse/doctor try reception tasks; as receptionist repeat.	Others blocked; receptionist allowed.	Blocked / as expected.	Passed	Screen rules match stored access rules.
Patient registration	New patient (form)	Receptionist; manual entry; fill required fields; submit.	Checks pass; duplicate check; patient saved; success message.	As expected,.	Passed	Gaps show clear errors.
Document upload	Allowed files	Upload PDF or image within size limit; check progress and preview.	Accepted; progress; preview; stored and linked to patient.	Upload, preview, link OK.	Passed	Only allowed types and sizes.
Document upload	Wrong file type	Upload a disallowed type (e.g. executable); read message.	Error shown; nothing stored.	Error; blocked.	Passed	No freeze or crash.
Reading documents	Text into form	Upload clear document; wait for reading; open review screen.	Fields filled; notes on trust; review opens.	Fields OK; review screen OK.	Passed	Bad scans need more editing.
Reading documents	Approve text	Review text; fix errors; approve to save.	Only approved data saved; patient and file linked; drafts do not live.	Review and save OK.	Passed	Drafts must not replace saved data.
Duplicate patients	Same details	Enter details like an existing patient; follow prompts to merge or cancel.	Warning before second record; staff can fix.	Warning OK; fix OK.	Passed	Rules and choices on screen.
Patient visits	Add visit	Receptionist; pick patient; enter visit type; submit.	Visit saved; linked; visible in history.	Saved; listed.	Passed	Matches hospital visit rules.
Activity history	View log	Allowed user; open log; filter by user, patient, or date if shown.	Who did what when; what filters work.	List and filters OK.	Passed	Supports accountability.

2.1.6 Daily Call Progress

Table 2.4 Standup meetings for Sprint 1

Date	Day	Sprint Phase	Tasks Discussed / Completed
Jan 1, 2026	Thu	Project kickoff	Sprint goals and backlog; chose web stack and cloud database with secure sign-in; repository and team rules.
Jan 2, 2026	Fri	Sign-in design	Started staff sign-in and passwords; draft patient storage design; workstations and project folders ready.
Jan 3, 2026	Sat	Registration screens	Sign-up screens and password checks; database changes applied; code review rhythm agreed.
Jan 4, 2026	Sun	Login screens	Login screens advanced; database updates tested; patient record shape begun; session handling next.
Jan 5, 2026	Mon	Login testing and roles	Login flows tested; staff roles and screen access defined; core patient record layout done.
Jan 7, 2026	Wed	Manual registration	Receptionist forms and field checks; new patients saved through the live system.
Jan 10, 2026	Sat	Document upload	PDF and image upload with type and size checks, progress, preview, secure storage, access rules.
Jan 13, 2026	Tue	Text extraction	Automatic text from documents into registration fields; notes on accuracy and errors.
Jan 16, 2026	Fri	Duplicate patients	Likely duplicates flagged before save; staff merge or dismiss workflow.
Jan 19, 2026	Mon	Review extracted data	Staff read, correct, and approve extracted text before it becomes official.
Jan 22, 2026	Thu	Records and permissions	Documents and extraction linked to patients; safe updates; access by job role.
Jan 24, 2026	Sat	Visits and activity log	Visit recording; who did what and when; simple filters.
Jan 27, 2026	Tue	Whole-path testing	Login through upload and extraction to approval; status during long steps.
Jan 29, 2026	Thu	Quality pass	Functional checklist: screens match roles; fixes on uploads and extraction.
Jan 31, 2026	Sat	Sprint closure	Review; stakeholder demo; documentation and architecture figure for Sprint 1 updated.

2.1.7 Committed Vs Completed User Stories

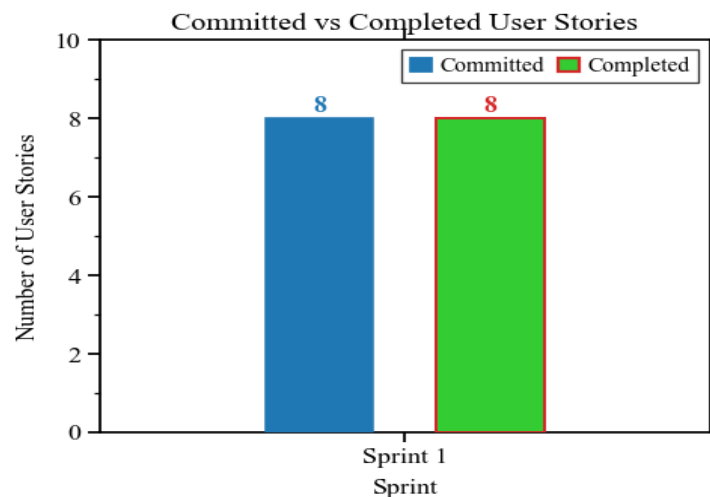


Figure 2.20 Bar graph for Committed Vs Completed User Stories for Sprint 1

2.1.8 Sprint Retrospective

Table 2.5 Sprint Retrospective for the Sprint 1

Liked	Learned	Lacked	Longed For
Clear ownership across the web app, sign-in, patient records, and document flows; steady teamwork.	How the browser app ties to hosted identity, database rules, storage, and review steps end to end.	Requirement shifts mid-sprint led to rework; edge cases not all nailed at kickoff.	Firm scope sign-off before sprint start and a light change-control rule.
Quick try-and-see cycles on uploads and registration; auth and role rules worked as designed.	Sound habits for file validation, user-visible errors, and careful database migrations.	Automatic text from scans struggled on handwriting; runs could feel slow; duplicate checks sometimes missed.	Other reading engines or background jobs for heavy pages; agreed speed and accuracy targets.
Coherent patient model; steady reviews; security foundation in good shape.	Realistic estimating for document reading; splitting big tasks and adding buffer.	Fewer automated tests than we wanted; some failures returned unclear messages.	Broader test coverage; consistent error reporting for faster fixes.
Scope delivered on time; foundation matches the plan.	Wiring approval so only checked data is saved; activity history for accountability.	Richer diagnostics when something fails in the pipeline.	Shared components for later sprints; a simple dashboard for onboarding health and audit views.

2.2 SPRINT 2

2.2.1 Sprint Goal with User Stories of Sprint 2

To build on Sprint 1 by delivering clinical workflows: role-appropriate patient lists and profiles (including assigned or active worklists where policy allows), append-only nursing and clinical documentation, QR identification at the bedside, doctor and intern notes with diagnoses, treatment, and prescriptions, case sheet assembly with print and controlled sharing where in scope, patient severity with intern access rules, and vitals with threshold alerts and monitoring with realtime updates in the UI. This sprint (Table 2.6) covers user stories US#9 through US#14 for the February–March 2026 increment.

Table 2.6 Detailed User Stories of sprint 2.

S. No	User Story
#US 9	As staff in each role, I want patient lists, profile access, and a My patients worklist governed by the same rules as screens and row-level data so that viewing versus editing stays consistent and I only see patients I am allowed to care for.
#US 10	As a nurse, I want nursing observations in an append-only medical record stream (no edit or delete of past rows), and as receptionist, doctor, intern, or nurse I want QR scan identification with safe failure on bad codes so that bedside verification is fast without exposing unnecessary patient detail.
#US 11	As a doctor or intern, I want progress notes, diagnoses, and treatment captured in an add-only way, plus prescription flows (drug, dose, route, duration) with visit and provider linkage, so that clinical documentation stays traceable and prescriptions respect access policy.
#US 12	As clinical staff, I want case sheets assembled from visits, diagnoses, prescriptions, and notes with a print-friendly view, reuse of document capture where OCR applies, and an optional minimal public case sheet link with create, renew, and cancel controls so that sharing is deliberate and compliant.
#US 13	As the institution, I want patient severity (Low, Medium, High) with interns blocked on High severity in session, navigation, and allocation, with clear access-denied messaging and badges on lists and headers so that policy is visible before opening a chart.
#US 14	As a nurse or clinician, I want vitals entry with consistent units, threshold-based alerts with sensible pacing where needed, monitoring dashboards, and realtime feeds so that abnormal values surface quickly and the UI stays usable under rapid updates.

The Planner Board representations of the user stories are shown in the figures below.

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SerenDoc

✓

US#9: Role Shaped Patient Access and Worklists

Completed on 04/03/2026 by SHYAM J (RA2212701010002)

SJ

PK

Authorization

×

Roles

×

Sprint 2

×

Bucket

Completed

▼

Progress

✓ Completed

▼

Priority

! Important

▼

Start date

01/02/2026

Due date

18/02/2026

Repeat

↻ Does not repeat

▼

Notes

☐ Show on card

- Nurses: worklist-oriented vitals, QR, and my patients; doctors and interns: profile, case sheet, prescriptions, allocation; reception and admin: broader lists where policy allows; **viewing vs. editing** in both screens and stored data
- **My patients**: assigned or active patients per rules; fresh lists when data changes; links to profile, case sheet, vitals; helpful empty states

Checklist 3 / 3

✓

Show on card

✓

Matrix of role × area × action; list vs. detail consistency for nurses vs. clinical staff

✓

Add-only history cannot be bypassed from alternate entry points; navigation matches role

✓

Rules for who appears on my-patients (active visit, allocation, filters)

Figure 2.21 User story: Role Shaped Patient Access and Worklists

SerenDoc

✓ US#10: Nursing Observations and QR Identification

Completed on 04/03/2026 by SHYAM J (RA2212701010002)



📁 Sprint 2 ✕ Clinical Notes ✕

Bucket	Progress	Priority
Completed ▾	✓ Completed ▾	! Important ▾
Start date	Due date	Repeat
05/02/2026 📅	26/02/2026 📅	↻ Does not repeat ▾

Notes

☐ Show on card

- **Nursing observation** entries in medical records; no edit or delete of past rows; single timeline with clear labels
- **QR scan** for receptionist, doctor, intern, nurse; shared interpretation; safe failure on bad codes; camera experience and security review (identifiers must not expose patient details)

Checklist 3 / 3

☒ Show on card

- ✓ Compact note entry; save type, author, time, and visit/patient link
- ✓ Block changes to older notes in both the screens and underlying services; timeline visibility
- ✓ Checks for reading QR codes; where each role lands after a good scan

Figure 2.22 User story: Nursing Observations and QR Identification

SerenDoc



US#11: Clinical Notes and Electronic Prescriptions

Completed on 04/03/2026 by SHYAM J (RA2212701010002)



AV

PK



Sprint 2

Clinical Notes

Bucket

Completed

Progress

Completed

Priority

Important

Start date

08/02/2026

Due date

25/02/2026

Repeat

Does not repeat

Notes

☐ Show on card

- Add-only **progress notes** and related information; intern rules by severity (see US#13); case sheet uses the same stream
- **Prescription** screens (including per-patient) for doctor and intern only; capture drug, dose, route, duration; access rules; case sheet link; print or export without leaking sensitive data

Checklist 3 / 3

☒ Show on card

- ☒ Diagnosis and treatment capture per agreed structure; add-only enforcement
- ☒ Link notes and prescriptions to visits and providers; intern visibility where severity restricts
- ☒ Prescription checks and warnings as scope allows

Figure 2.23 User story: Clinical Notes and Electronic Prescriptions

SerenDoc



US#12: Case Sheets and Controlled Public Sharing

Completed on 04/03/2026 by SHYAM J (RA2212701010002)



SJ

PK



OCR

Sprint 2

Clinical Notes

Bucket

Completed

Progress

Completed

Priority

Important

Start date

12/02/2026

Due date

03/03/2026

Repeat

Does not repeat

Notes

☐ Show on card

- Case sheet from visits, diagnoses, prescriptions, notes; print-friendly; document capture for case uploads; human review when confidence is low
- **Public** view: smallest necessary field set; create, renew, and cancel share access; limits and abuse resistance; compliance sign-off

Checklist 2 / 2

☒ Show on card

- ☒ Staff-only assembly; print without sensitive data in logs; reuse document capture with a review...
- ☒ Minimize public fields; audit share access if required

Figure 2.24 User story: Case Sheets and Controlled Public Sharing

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SerenDoc

✓ **US#13: Severity-Based Access Constraints**

Completed on 04/03/2026 by SHYAM J (RA2212701010002)

AV

SJ

🏠

Sprint 2 ✕

Access control ✕

Bucket

Completed ▾

Progress

✓ Completed ▾

Priority

! Important ▾

Start date

15/02/2026 📅

Due date

05/03/2026 📅

Repeat

🔄 Does not repeat ▾

Notes

☐ Show on card

- Interns blocked on **high** severity (session, allocation, navigation); clear access-denied messaging; badges on lists and headers

Checklist 2 / 2

✓

Show on card

✓

Store severity; align with vitals where applicable; doctor access per policy

✓

Role × severity verification

Figure 2.25 User story: Severity Based Access Constraints

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SerenDoc

✓

US#14: Vitals Lifecycle: Capture, Alerts, Dashboards, and Realtime

Completed on 1 days ago by SHYAM J (RA2212701010002)

👤

AV

PK

📅

Sprint 2

×

📅

Vitals

×

Bucket

Completed

▼

Progress

✓ Completed

▼

Priority

! Important

▼

Start date

20/02/2026

📅

Due date

15/03/2026

📅

Repeat

🔄 Does not repeat

▼

Notes

☐ Show on card

- Units and fields consistent; threshold warnings and criticals; optional pause between repeated alerts; **live updates** where enabled; simulated streams for demos; jump from abnormal vitals to patient context; performance when updates arrive quickly

Checklist 3 / 3

✓

Show on card

✓

Threshold evaluation and when to raise alerts; screens receive live vitals updates where used

✓

Charts and summary cards with accessible contrast; empty and error states

✓

Verification under rapid updates; tune thresholds per environment

⌋

...

⌋

Figure 2.26 User story: Vitals Lifecycle: Capture, Alerts, Dashboards, and Realtime

2.2.2 Functional Document

2.2.2.1. Introduction

This document outlines functionality delivered in Sprint 2 of SerenDoc. The sprint adds clinical workflows on top of Sprint 1: role-appropriate access to patient lists and records, append-only nursing and clinical notes, case sheet surfaces and OCR-assisted intake where applicable, severity rules for interns, and vitals with alerts and realtime updates. Staff use the same web stack as Sprint 1—Supabase (PostgreSQL, Storage, Realtime, Edge Functions) and a React client—with nursing and doctor or intern entries appearing in one timeline where the product distinguishes note types.

2.2.2.2. Product Goal

The primary goal is to give nurses and clinicians structured, trustworthy workflows on top of Sprint 1 data: who may see which patients, what each role may add without rewriting history, how case sheets are composed and shared, how severity gates intern access, and how vitals and alerts behave in the UI. The platform aims to:

- Enforce read versus write and append-only rules for clinical and nursing content at both UI and service layers.
- Support nursing observations, diagnoses, treatment plans, progress notes, and prescriptions with linkage to patients and visits.
- Provide case sheet assembly and print, with OCR reuse and human review when extraction confidence is low.
- Apply severity consistently with intern limits and clear access-denied experiences.
- Deliver vitals display and entry paths, threshold alerts, dashboards, and realtime updates.

2.2.2.3. Demography (Users, Location)

Users:

- Nurses: vitals- and worklist-oriented flows; nursing observations; QR; view history; no edits to past rows.
- Doctors and interns: case sheet, clinical notes, diagnoses, treatment, prescriptions; interns constrained by severity policy.
- Receptionists: Sprint 1 onboarding capabilities; patient lists where policy allows; case sheet upload or facilitation as designed.
- Administrators: broader lists or oversight where product policy places them (aligned with planner role areas).

Location:

- Hospitals and similar facilities (single or multi-site)

2.2.2.4. Business Processes

- Staff open patient lists and profiles filtered by role; My patients shows assigned or active patients per rules, with links to profile, case sheet, vitals, and notes.
- Nurses add observations; doctors and interns add diagnoses, treatment, and progress notes; all historical clinical rows remain immutable.
- QR scans resolve to the correct patient context for allowed roles; invalid or unsafe codes fail without leaking identifiers.

- Prescriptions are captured for authorised roles, linked to visits and providers, with export or print paths that avoid casual leakage of sensitive data.
- Case sheets aggregate structured clinical data; optional public links expose the smallest necessary field set with lifecycle control (create, renew, cancel).
- Severity is stored and shown; interns cannot work High severity cases; doctors can, per policy.
- Vitals feed dashboards and may drive severity updates where designed; alerts fire on thresholds with optional debouncing in busy environments.

2.2.2.5. Features

This project focuses on implementing the following key features:

Feature 1: Role-shaped patient access and worklists

1. Description

- Lists, detail screens, and APIs enforce the same role-by-action rules; append-only history cannot be bypassed from alternate entry points.
- My patients and filters reflect active visit, allocation, or assignment rules with helpful empty states.

2. User story

- As a nurse, doctor, intern, or receptionist, I want list and detail behaviour that matches my role and My patients rules so that I only act on patients I am allowed to care for.

Feature 2: Nursing observations, QR, and clinical documentation

1. Description

- Nursing observations append to the medical record stream with author, time, type, and visit/patient linkage; past rows are not editable.
- QR scanning supports bedside identification for allowed roles with shared interpretation of codes and safe failure modes.

2. User story

- As a nurse, I want to add observations that stay in an immutable timeline, and as clinical staff I want QR identification that lands me in the right patient context without exposing unnecessary detail.

Feature 3: Clinical notes, prescriptions, case sheets, severity, and vitals

1. Description

- Doctors and interns add diagnoses, treatment, progress notes, and prescriptions under append-only and severity rules.
- Case sheets combine visits, diagnoses, prescriptions, and notes; print and optional public link flows include compliance-oriented minimisation of fields.
- Vitals use consistent units; dashboards and alerts support realtime updates.

2. User story

- As a doctor or intern, I want complete clinical documentation and prescriptions within policy; as any authorised role, I want case sheets and vitals that reflect severity and alerts without breaking auditability.

2.2.2.6. Authorisation Matrix

Table 2.7 Access level Authorisation Matrix of sprint 2

User role	Lists & profiles	Clinical documentation	Case sheets & vitals
Receptionist	Yes	No	Yes
Administrator	Yes	No	Yes
Nurse	Yes	Yes	Yes
Doctor	Yes	Yes	Yes
Intern	Yes	Yes	Yes
Unauthenticated	No	No	No

2.2.2.7. Assumptions

- Sprint 1 foundations (identity, RLS, storage, OCR pipeline) remain available and stable.
- Vitals sources and device bindings follow the deployment configuration; behaviour is validated in the target environment.
- OCR on handwritten case sheets may be slower or less accurate; low-confidence extractions require human review before auto-append where required.
- Public case sheet links, if enabled, undergo organisational compliance review; fields are minimised by design.
- Sprint 2 scope is fixed for the reporting baseline: Sprint 3 builds intelligence, allocation, audit, and dashboards on this layer.

2.2.3 Architecture Document

2.2.3.1 Application

SerenDoc in Sprint 2 keeps the Supabase-centric architecture from Sprint 1 and adds clear modules for clinical access, append-only medical records (nursing versus progress notes), prescriptions, case sheet assembly and optional public sharing, severity enforcement, QR resolution, and vitals with alerts and realtime updates. The React client continues to use the Supabase client for Postgres, Storage, Realtime, and Edge Functions; route and row policies align role, severity, and visit context.

Key logical modules include:

- Clinical access service — Role-shaped lists, profile routes, My patients, and read/write matrices.
- Nursing documentation service — Append-only observations linked to visits and patients.
- Clinical documentation service — Diagnoses, treatment, progress notes; intern gating by severity.
- Prescription service — Structured drug, dose, route, duration; linkage to visits and providers.
- Case sheet service — Aggregation, print views, OCR-assisted document path, optional tokenised public view.
- Severity service — Low, Medium, and High storage, badges, and intern restriction checks.
- QR identification service — Parse and validate codes; role-appropriate navigation targets.
- Vitals and alerting service — Schema, thresholds, dashboard feeds, realtime ingestion paths.

2.2.3.2 System Architecture

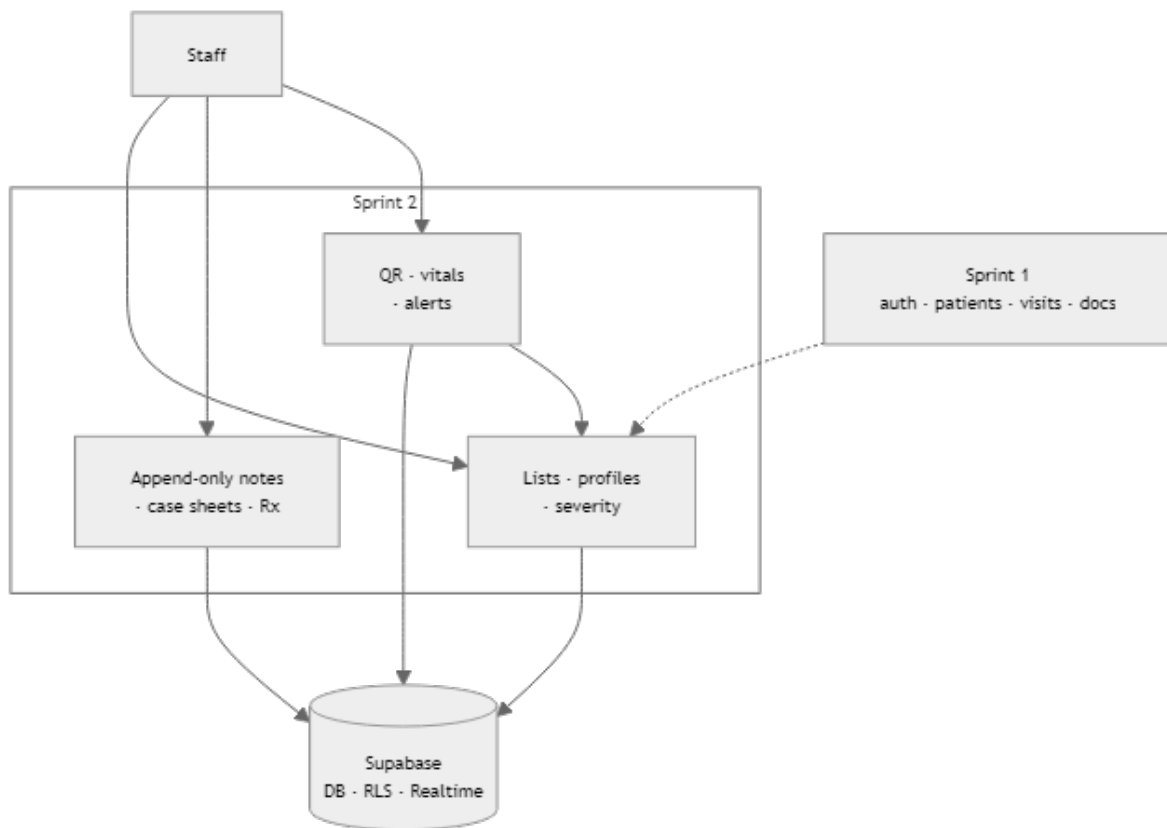


Figure 2.27: System Architecture Diagram (Sprint 2)

Summary: The figure shows the Sprint 2 layer only: role-shaped lists and profiles, severity rules, append-only clinical documentation (nursing and doctor or intern), case sheets, QR identification, and vitals with alerts, all tied to Supabase Postgres and Realtime. Sprint 1 foundations (identity, patient master, visits, documents) are not redrawn here; they appear in the Sprint 1 architecture figure.

2.2.3.3 Data Exchange Contract

Frequency of data exchanges

- **Interactive:** Patient list and profile loads, note and prescription saves, case sheet render, QR lookup, severity checks, vitals entry.
- **Streaming / high-churn:** Vitals updates and alert notifications via Realtime or equivalent subscriptions where configured.
- **Asynchronous:** OCR on case sheet documents may run in Edge Functions with status via Realtime or polling; long scans should not freeze the UI.

Data sets

- **Staff and role context** — Session, role, severity eligibility for interns, and allocation hints where used.
- **Patient and visit data** — As in Sprint 1, plus clinical append-only rows, prescriptions, severity, vitals time series.
- **Case sheet and sharing data** — Aggregated clinical snapshot, print layout, optional public token metadata and expiry.
- **QR payload data** — Opaque or scoped identifiers that resolve server-side without exposing unnecessary PHI in the code itself.
- **Vitals and alert data** — Measurements, thresholds, alert events, optional link to severity updates.

Mode of exchanges

- **Supabase REST / RPC** — Authoritative reads and transactional writes with RLS and severity checks.
- **Supabase Realtime** — Vitals streams and alert fan-out where enabled.
- **Supabase Storage** — Case sheet uploads when documents are ingested through the file pipeline.
- **Edge Functions** — OCR or heavy validation steps consistent with Sprint 1 patterns.

2.2.4 UI Design

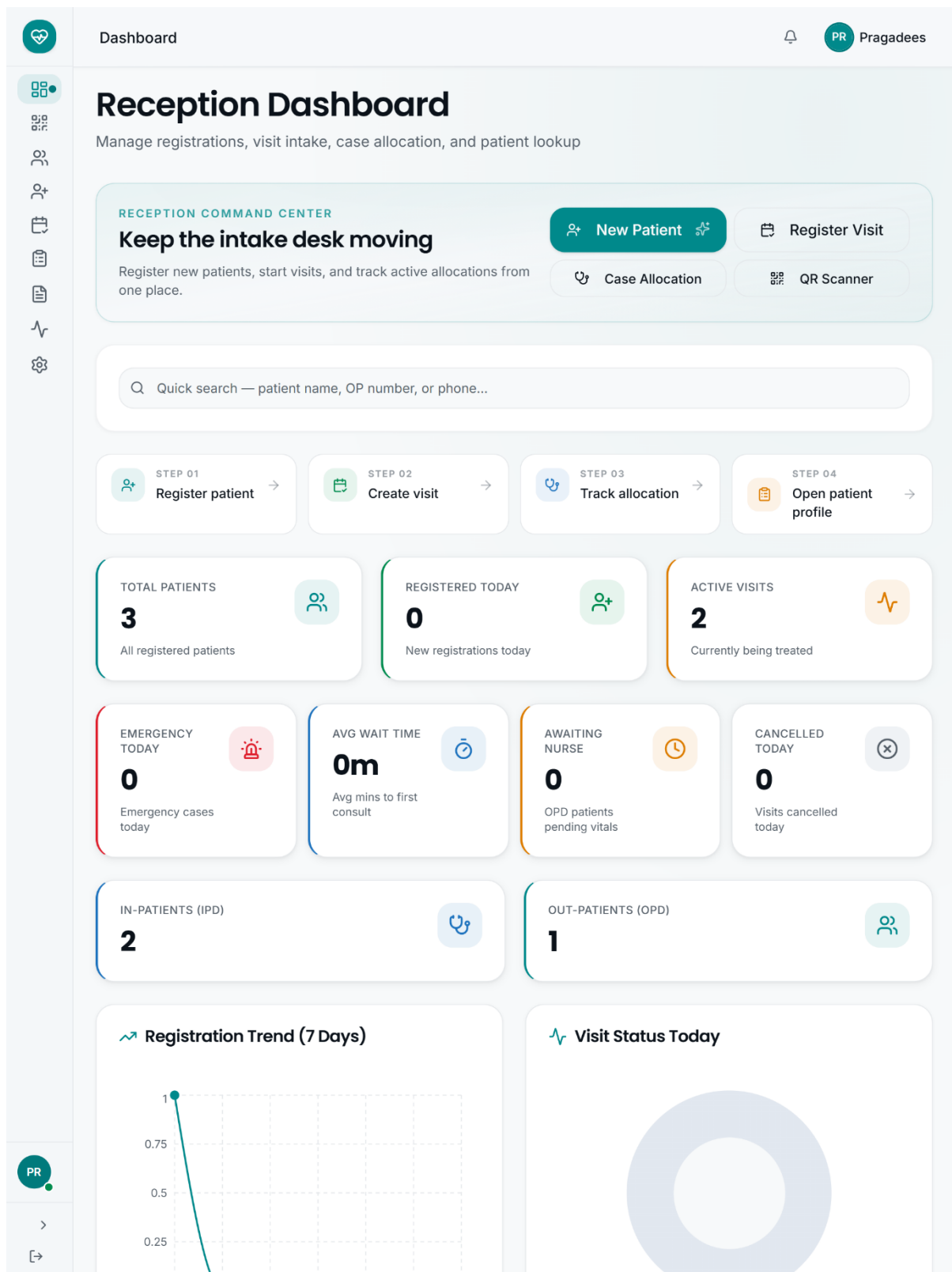


Figure 2.28 UI Design for Receptionist Dashboard Page

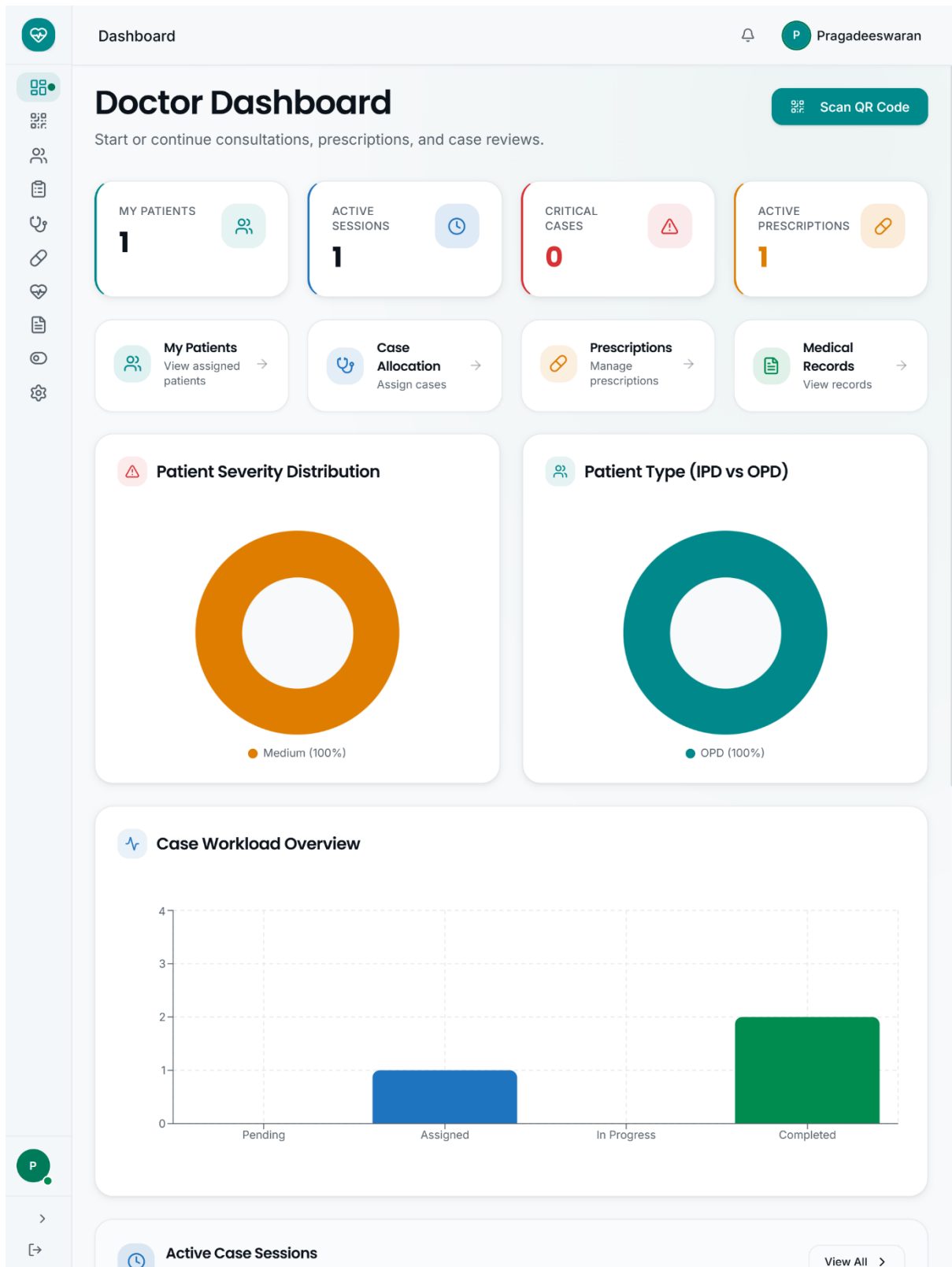


Figure 2.29 UI Design for Doctor Dashboard Page

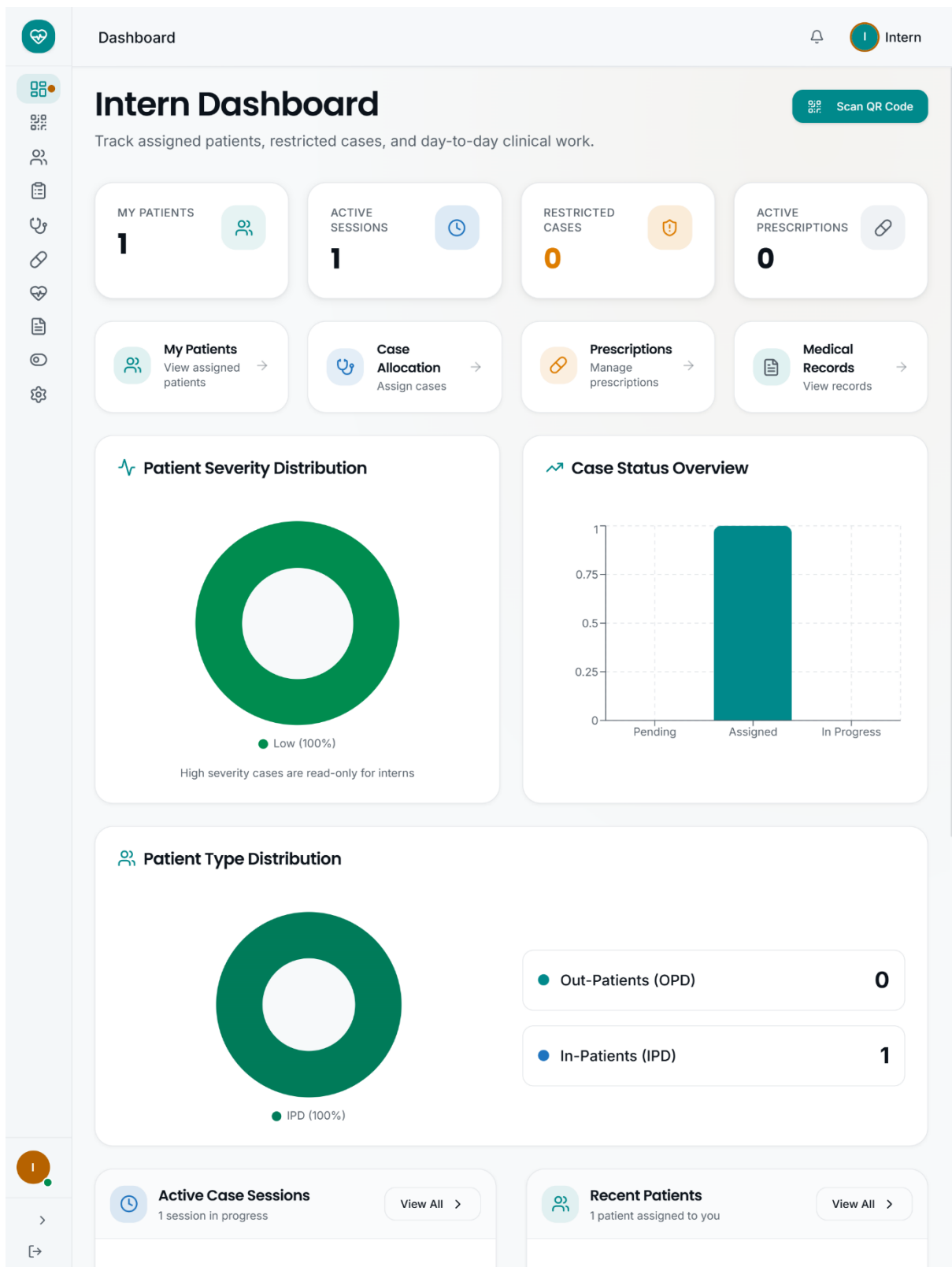
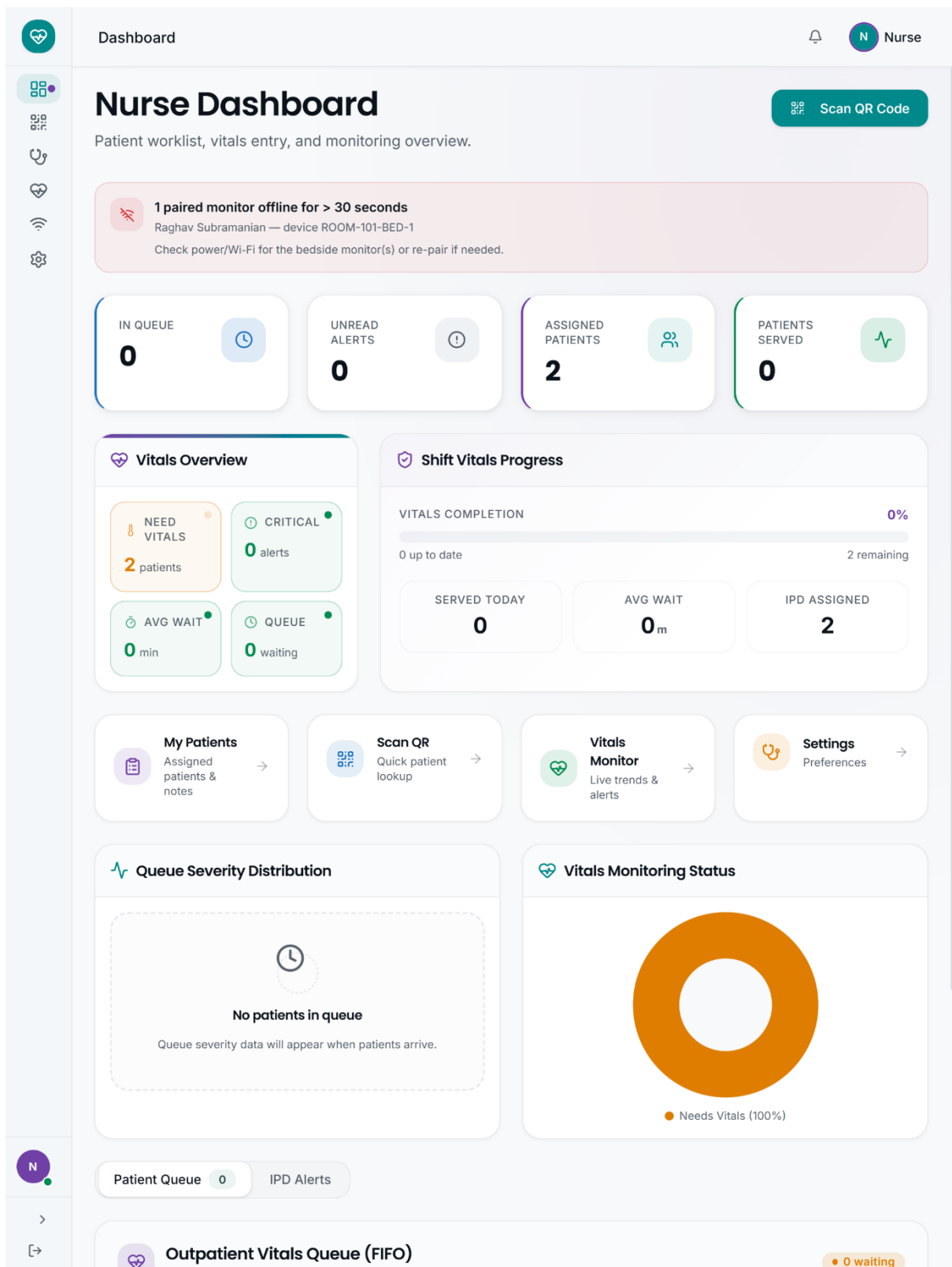


Figure 2.30 UI Design for Intern Dashboard Page



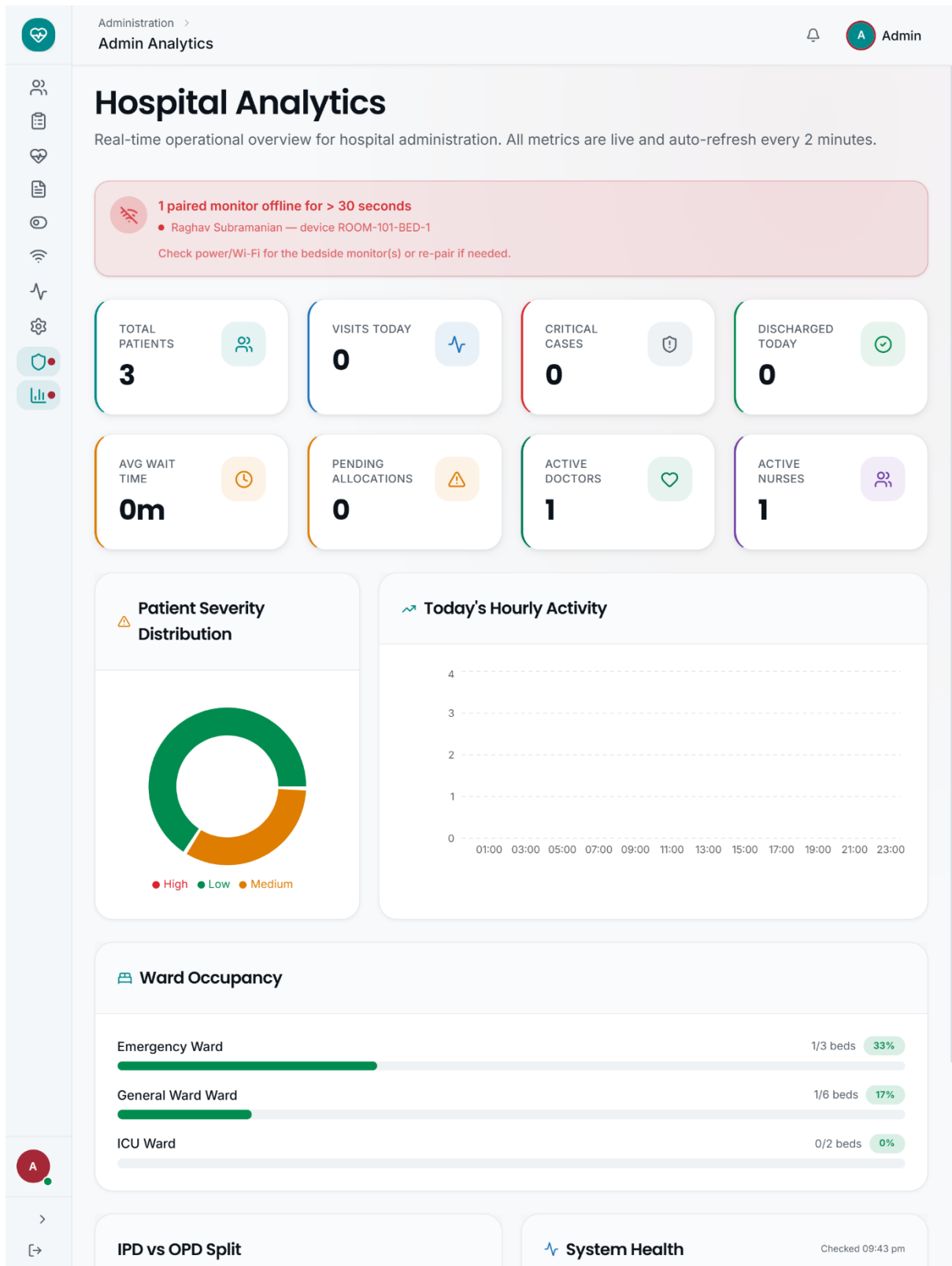
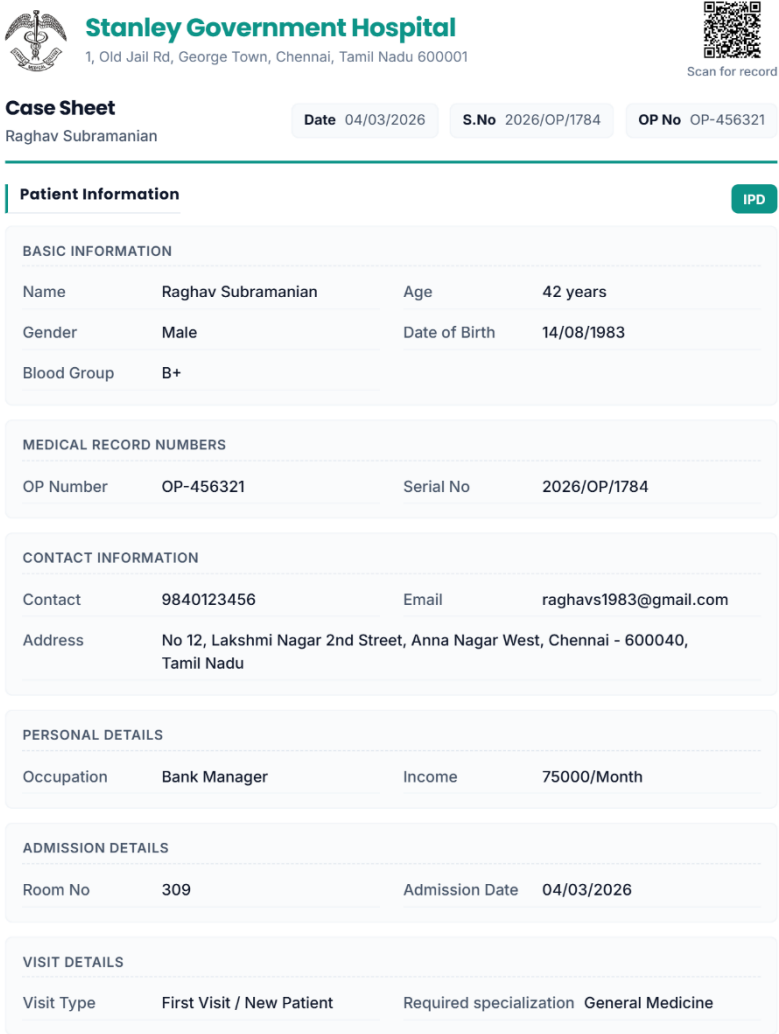


Figure 2.32 UI Design for Admin Dashboard Page (Admin Analytics)



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2.2.5 Functional Test Cases

Table 2.8 Detailed Functional Test Case for Sprint 2

Feature	Test Case	Steps to Execute Test Case	Expected Output	Actual Output	Status	More Information
Role access & worklists	Lists, profiles, My patients	Open lists and profile as nurse, doctor, intern; use My patients where applicable; confirm viewing vs editing matches role.	Same rules in UI and stored data; worklist shows allowed patients only.	As expected.	Passed	Role × screen × row checks.
Nursing & QR	Append-only nursing; QR to patient	Nurse adds observation; try to edit past row; scan QR as allowed role; bad code fails safely.	New nursing row only; past immutable; QR lands in correct patient context without leaking identifiers.	As expected.	Passed	Single timeline; QR failure modes reviewed.
Clinical notes & Rx	Diagnosis, treatment, progress, prescription	Doctor/intern add clinical entries and prescription fields; try to overwrite prior row.	Append-only rows; prescriptions linked to visit/patient; intern rules apply.	As expected.	Passed	Matches add-only clinical path.
Case sheet	Assemble, print, OCR path	Open case sheet from visits/notes/Rx; upload document OCR path if used; optional public link controls if in scope.	Printable assembly; OCR append after review where required.	In scope per Sprint 2 definition.	Passed	Low-confidence OCR needs human confirm.
Severity	Intern blocked on High	Set patient to High; intern opens; doctor opens; check badges on list/header.	Intern denied or redirected; doctor allowed; severity visible before chart open.	As expected.	Passed	Policy visible in navigation.
Vitals	Realtime feed and threshold alert	Open vitals view; observe updates; drive value across threshold.	Updates without full page refresh; alert when threshold crossed.	Realtime/streaming OK; alert logic OK.	Passed	Thresholds environment specific.

Table 2.8 shows function test case of sprint 2

2.2.6 Daily Call Progress

Table 2.9 Standup meetings for Sprint 2

Date	Day	Sprint Phase	Tasks Discussed / Completed
Feb 1, 2026	Sun	Sprint 2 kickoff	Scope for clinical workflows; role-based lists and profiles; append-only rules agreed.
Feb 4, 2026	Wed	Access and nursing design	View permissions and API protection; nursing notes wireframes; edit restrictions for past rows.
Feb 8, 2026	Sun	Nursing complete; clinical start	Nursing workflow tests passed; clinical notes and diagnosis work begun.
Feb 11, 2026	Wed	Clinical notes advance	Treatment and progress notes; append-only clinical writes; case sheet OCR exploration.
Feb 14, 2026	Sat	Case sheets and severity	Case sheet upload and OCR integration; severity Low/Medium/High design; intern rules.
Feb 17, 2026	Tue	US#12 and US#13 sign-off themes	Case sheet OCR slice complete; severity UI and access tests with role matrix.
Feb 20, 2026	Fri	Vitals and alerts	Vitals model; threshold alerts; dashboard integration started.
Feb 24, 2026	Tue	Vitals sign-off	Vitals flows signed off; cross-feature testing (access, severity, vitals).
Feb 28, 2026	Sat	Regression buffer	Development complete; final verification; closure week preparation.
Mar 3, 2026	Tue	Stakeholder demo	Demo completed; backlog feedback captured.
Mar 5, 2026	Thu	Retrospective and close	Sprint 2 retrospective; sprint closed; handover to Sprint 3.

Table 2.9 Shows the standup meeting of sprint 2

2.2.7 Committed Vs Completed User Stories

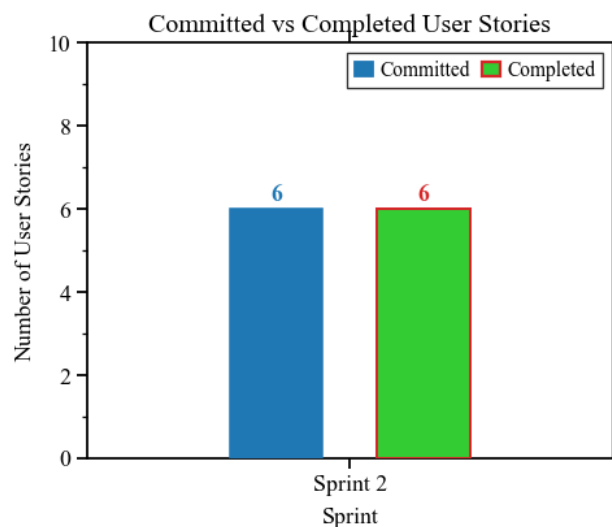


Figure 2.34 Bar graph for Committed Vs Completed User Stories for Sprint 2

2.2.8 Sprint Retrospective

Table 2.10 Sprint Retrospective for the Sprint 2

Liked	Learned	Lacked	Longed For
All sprint user stories delivered; append-only nursing and clinical paths behaved as designed.	How severity, vitals, and role gates interact end to end in the same patient journey.	Handwritten case sheet OCR could be slow and variable; some alert noise in dev.	Faster or specialised handwriting OCR; mandatory review when confidence is low; calmer alert thresholds.
Vitals and alerts integrated; realtime updates worked in testing.	Clear access-denied copy reduces support confusion when interns hit High severity.	Role × severity edge cases (e.g. mid-session changes) not fully exhausted.	Full role-by-severity-by-action test matrix; standardized API errors for 403 and severity denials.
Code reviews stayed consistent; documentation matched clinical workflow intent.	Importance of consistent vitals units and dashboard performance under rapid updates.	Inconsistent error shapes across APIs for access failures.	Single error contract for auth/severity; configurable alert debounces per environment.
Strong foundation for Sprint 3 (allocation, central severity service, audit dashboards).	Balancing realtime vitals load with stable dashboard behaviour.	Broader automated coverage for access rules.	Shared UI patterns for badges, tooltips, and denial screens across roles.

2.3 Sprint 3

2.3.1 Sprint Goal with User Stories of Sprint 3

Sprint 3 adds governance on top of earlier work: vitals-driven severity with a fallback when the service is down, case allocation under severity and role rules, audit and operational activity views, role and admin dashboards with IoT oversight where built, patient and visit status changes, and release hardening for quality, security, and demo. Table 2.11 lists user stories US#15–US#20 (March–April 2026).

Table 2.11 Detailed User Stories of sprint 3.

S. No	User Story
#US 15	As the platform, I want a central severity classification service fed by vitals with a backup calculation path when that service is unavailable, plus severity-change notifications and clear provenance of results, so that patient severity stays consistent without exposing sensitive health data in logs.
#US 16	As care coordination staff, I want case allocation that considers severity, specialization, workload, and intern versus doctor rules, with automatic assignment where enabled, manual override, and safe behaviour under concurrency, so that patients are matched to appropriate providers.
#US 17	As the institution, I want a tamper-evident audit trail with who, when, what, and record identifiers, and a separate operational activity timeline for reception and administration, so that governance and day-to-day operations are both visible without mixing forensic audit with routine activity.
#US 18	As staff and administrators, I want role dashboards, an admin home with analytics, IoT fleet views for authorised roles, and safe settings for signed-in users, so that each role lands on relevant metrics and device health without leaking unnecessary data.
#US 19	As a doctor, intern, or administrator, I want controlled patient and visit status transitions that align with severity and write audit entries, so that workflow state stays coherent under concurrent updates.
#US 20	As the delivery team, I want automated checks, security validation, accessibility on primary flows, and a documented happy path for demonstration, so that release quality and stakeholder demos are repeatable without sensitive data in artefacts.

The Planner Board representations of the user stories are shown in the figures below.

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SerenDoc

✓

US#15: ~~Severity Classification from Vitals~~

Completed on 1 days ago by SHYAM J (RA2212701010002)

SJ

PK

Sprint 3

×

AI&ML

×

Bucket

Completed

▼

Progress

✓ Completed

▼

Priority

! Important

▼

Start date

01/03/2026

Due date

20/03/2026

Repeat

↺ Does not repeat

▼

Notes

☐ Show on card

- Organization service as source of truth; **backup calculation** in the app when the service fails; outputs low, medium, or high; **severity-change** notifications; record whether the result came from the main service or the backup, without logging sensitive health information

Checklist 2 / 2

☐ Show on card

✓ Agreed service behavior and updates; align backup behavior with service rules

✓ Boundary checks for representative vital sign sets

Figure 2.35 User Story: Severity Classification from Vitals

SerenDoc

✓ US#16: Case Allocation Engine

Completed on 1 days ago by SHYAM J (RA2212701010002)



🏠 Sprint 3 × Automation and allocation ×

Bucket	Progress	Priority
Completed ▾	✓ Completed ▾	! Important ▾
Start date	Due date	Repeat
05/03/2026 📅	25/03/2026 📅	🔄 Does not repeat ▾

Notes

☐ Show on card

- Provider selection rules; automatic assignment after vitals where enabled; manual override; safe retries; verification when many allocations run at the same time

Checklist 2 / 2

☒ Show on card

- ✓ Document priority rules and tie-breakers; clear outcomes when allocation cannot complete
- ✓ Show outcomes on dashboards and notifications

Figure 2.36 User Story: Case Allocation Engine

SerenDoc

✓ US#17: Audit Trail and Operational Activity

Completed on 1 days ago by SHYAM J (RA2212701010002)



📁 Sprint 3 ✕

Bucket

Completed



Progress

✓ Completed



Priority

! Important



Start date

10/03/2026



Due date

05/04/2026



Repeat

↺ Does not repeat



Notes

☐ Show on card

- **Audit:** who, when, what, record identifiers, snapshots; add-only storage; viewer with filters; audit records cannot be changed after they are written
- **Activity** for receptionist and administrator: operational "what changed today?" — different from forensic audit; long lists handled smoothly; consistent date and time display

Checklist 3 / 3

☒ Show on card

- ✓ Audit event types and redaction; cover registration, notes, allocation, severity, status
- ✓ Filters when reviewing activity; which events are activity vs. audit-only
- ✓ Accessibility for long timelines

Figure 2.37 User Story: Audit Trail and Operational Activity

SerenDoc

✓ US#18: ~~Dashboards, Admin Oversight, IoT Fleet, and Settings~~

Completed on 1 days ago by SHYAM J (RA2212701010002)



Sprint 3 X Dashboard & UI/UX X

Bucket	Progress	Priority
Completed	✓ Completed	! Important
Start date	Due date	Repeat
12/03/2026	10/04/2026	Does not repeat

Notes

☐ Show on card

- **Default dashboard** → role-specific content; **Administrator** → **Admin home**; quick answers for each role; alerts and severity badges
- **Admin analytics**: KPI widgets, prepared summaries, admin-only access
- **IoT fleet**: nurse and administrator; device list, status, offline or stale alerts; steps to add or remove devices
- **Settings**: all signed-in roles; safe profile subset, notification toggles; no sensitive health data in optional fields

Checklist 4 / 4

☐ Show on card

- ✓ Smoke checks for receptionist, nurse, doctor/intern, and admin dashboards; loading states
- ✓ KPI definitions and backing data; export or scheduled reports if in scope
- ✓ IoT drill-down to patient/device linkage
- ✓ Settings accessibility and mobile layout

Figure 2.38 User Story: Dashboards, Admin Oversight, Iot Fleet, and Settings

SerenDoc



US#19: Patient and Visit Status Management

Completed on 1 days ago by SHYAM J (RA2212701010002)



AV

SJ



Vitals

Sprint 3

Dashboard & UI/UX

Bucket

Completed

Progress

Completed

Priority

Important

Start date

18/03/2026

Due date

05/04/2026

Repeat

Does not repeat

Notes

☐ Show on card

- States (for example under treatment, critical, discharged) per policy; align with severity (US#13, US#15); audit entries (US#17); concurrent update edge cases

Checklist 1 / 1

☒ Show on card

- List transitions and who may perform them; reflect on dashboards and lists

Figure 2.39 User Story: Patient and Visit Status Management

... X

SerenDoc

○ US#20: System Hardening and Final Demo






Bucket Backlog ▾	Progress ☐ Not started ▾	Priority  Important ▾
Start date 01/04/2026 	Due date 30/04/2026 	Repeat  Does not repeat ▾

Notes ☐ Show on card

- Automated checks at multiple levels; quality checks before release; safe browsing protections and third-party package review; no secrets in shared materials; documented happy path: register → visit → vitals → alerts → audit/activity

Checklist 0 / 3 ☐ Show on card

- ☐ End-to-end smoke and regression (no sensitive data in recordings); accessibility on primary flo...
- ☐ Security checklist: sign-in, who may see which records, system touchpoints, share and access c...
- ☐ Performance on dashboards; demo materials and frozen demo environment

Figure 2.40 User Story: System Hardening and Final Demo

2.3.2 Functional Document

2.3.2.1. Introduction

This document describes functionality delivered in Sprint 3 of SerenDoc. The sprint builds on Sprint 1 (onboarding and identity) and Sprint 2 (clinical workflows) by adding vitals-informed severity with service and fallback paths, allocation engines, audit and activity surfaces, role and administrator dashboards, IoT fleet oversight where implemented, visit and patient status management, and release hardening. The stack remains Supabase (PostgreSQL, Storage, Realtime, Edge Functions) with a React web client.

2.3.2.2. Product Goal

The primary goal is to make severity, assignment, visibility, and traceability operational at scale: severity reflects vitals and policy; allocation respects roles and workload; dashboards give each role actionable landings; audit and activity explain what changed; status moves are controlled; and the product is fit for demonstration and review. The platform aims to:

- Classify severity from vitals with server-led rules and a defined client fallback when needed.
- Allocate cases using severity, specialization, workload, and visit context, with clear outcomes when allocation cannot complete.
- Record append-only audit events and present operational activity without rewriting history.
- Deliver role dashboards, administrator analytics, fleet views for authorised roles, and safe settings.
- Manage patient and visit status with policy-aligned transitions and audit coverage.
- Close the release with linting, tests, security checks, and an agreed end-to-end rehearsal path.

2.3.2.3. Demography

Users

- Receptionists: dashboards, intake and visit flows carried forward, case allocation, activity views, settings as allowed.
- Nurses: dashboards, worklists, vitals, QR, IoT fleet where policy allows, settings.
- Doctors and interns: dashboards, allocation, My patients, vitals, records, status, prescriptions, QR, case sheets, settings; interns remain subject to severity rules from earlier sprints.
- Administrators: admin home, analytics, fleet, activity, status, patients and profiles as permitted; not registration or visit intake where those are reception-only.

Location

- Hospitals and similar facilities (single or multi-site), consistent with prior sprints.

2.3.2.4. Business Processes

- Vitals feed severity classification; the application records whether results came from the primary service or the fallback path.

- Allocation selects providers using agreed priority and tie-breakers; high-severity auto-assignment excludes interns where rules require; concurrent attempts surface clear errors.
- Mutating actions append audit rows; activity lists combine audit and operational entries for reception and administration with filters and stable date and time display.
- Default routes open role dashboards; administrators use admin and analytics pages for KPI-style summaries.
- IoT fleet pages list devices, status, and stale or offline signals for nurse and administrator roles where built.
- Status transitions follow policy, align with severity where applicable, and leave audit evidence.
- Hardening runs quality and security checks; demo materials avoid real patient identifiers.

2.3.2.5. Features

Feature 1: Severity, allocation, and status

1. Description

- Central severity from vitals with backup behaviour when the service is unavailable, notifications on material changes.
- Case allocation with manual and automatic paths, workload and specialization awareness, and status transitions for authorised roles.

2. User story

- As care staff and administrators, I want severity, allocation, and visit or patient status to follow one set of rules and to show up consistently on dashboards and lists.

Feature 2: Audit, activity, and governance

1. Description

- Append-only audit storage with viewer filters, operational activity distinct from forensic audit for reception and administration.

2. User story

- As reception and administration, I want trustworthy audit and activity views so that we can explain what changed and when during operations and review.

Feature 3: Dashboards, fleet, settings, and hardening

1. Description

- Role dashboards and administrator analytics; IoT fleet oversight for authorised roles; profile and notification settings without optional sensitive health fields.
- Automated checks, security validation, accessibility on primary flows, and a documented demo path.

2. User story

- As each role, I want a relevant landing experience and, where applicable, fleet and admin insight; as an organisation, I want the release proven by checks and a safe demo script.

2.3.2.6. Authorisation Matrix

Table 2.12 Access level Authorisation Matrix of sprint 3

User role	Dashboards & settings	Allocation & visit status	Audit & activity
Receptionist	Yes	Yes	Yes
Nurse	Yes	No	Yes
Doctor	Yes	Yes	Yes
Intern	Yes	Yes	No
Administrator	Yes	Yes	Yes
Unauthenticated	No	No	No

Table 2.12 shows the access level authorisation matrix for Sprint 3. Allocation is led by reception and coordination roles per product rules; audit and activity detail varies by role; intern access follows severity policy from earlier sprints.

2.3.2.7. Assumptions

- Sprint 1 and Sprint 2 capabilities remain available; database, Storage, and Realtime meet load for dashboards and audit volume in the target environment.
- Severity service endpoints and fallback behaviour are configured for staging and production; boundary tests cover representative vitals sets.
- Demonstration and test data contain no real patient identifiers.

- Sprint 3 scope is fixed for this report baseline; further product ideas stay in the backlog, not in this write-up.

2.3.3 Architecture Document

2.3.3.1 Application

SerenDoc in Sprint 3 extends the Supabase-backed platform with services for severity classification (primary and fallback), allocation orchestration, append-only audit persistence, operational activity presentation, role and administrator dashboards, IoT fleet listing and health signals, status transition handling, and hardening hooks (CI, security checks). The **React** client consumes Supabase APIs and Realtime channels for live dashboards and activity feeds.

Key logical modules include:

- Severity service — Vitals inputs, rule evaluation, fallback path, notifications, provenance flags.
- Allocation service — Provider selection, workload and specialization inputs, intern exclusion rules, concurrency handling.
- Audit and activity service — Append-only audit writes, merged or filtered activity views, redaction rules.
- Dashboard and analytics service — Role payloads, administrator KPIs, indexed aggregates where used.
- Fleet service — Device registry, connectivity or staleness signals for authorised roles.
- Status service — Allowed transitions, validation against severity and visit context, audit emission.
- Hardening — Lint, tests, accessibility checks, release and demo documentation.

2.3.3.2 System Architecture

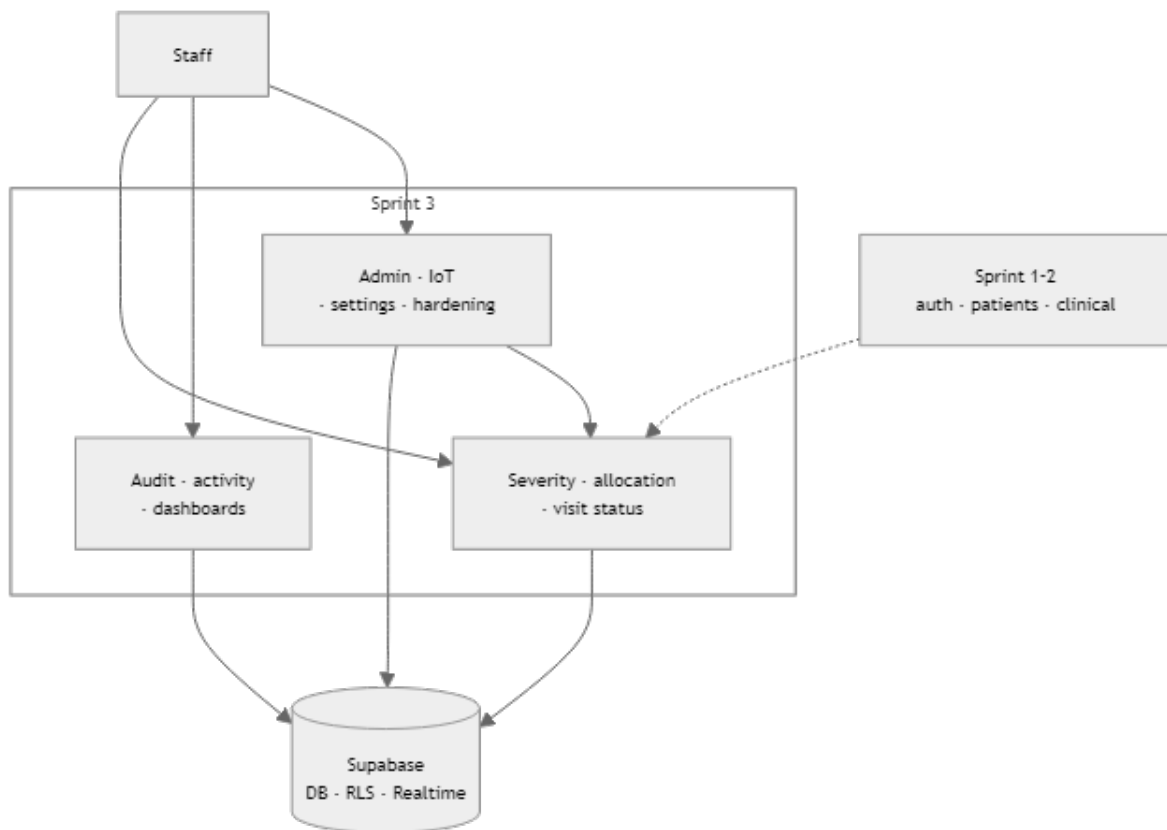


Figure 2.41 System Architecture Diagram (Sprint 3)

2.3.3.3 Data Exchange Contract

Frequency of data exchanges

- **Interactive:** Dashboard loads, allocation actions, status changes, settings saves, audit queries.
- **Streaming / high-churn:** Realtime updates for activity lists, vitals or fleet signals where subscribed.
- **Batch or scheduled:** Optional KPI refresh or rollups if implemented for analytics widgets.

Data sets

- **Severity results** — Level, source (service or fallback), timestamps, linkage to patient or visit.
- **Allocation data** — Assignee, visit, rationale fields, override flags, concurrency metadata.

- **Audit and activity rows** — Actor, entity, action, snapshots where allowed, operational versus audit classification.
- **Dashboard and KPI data** — Aggregates and role-specific card payloads.
- **Fleet data** — Device identifiers, status, last seen, linkage rules where stored.
- **Status data** — Current state, transition history references, alignment with severity where required.

Mode of exchanges

- **Supabase REST / RPC** — Reads and transactional writes with RLS.
- **Supabase Realtime** — Live activity and monitoring subscriptions where enabled.
- **Edge Functions** — Severity or allocation side-effects, audit fan-out, or webhooks as configured.
- **PostgreSQL** — Append-only audit tables, transactional status updates, indexed queries for analytics.

2.3.4 UI Design

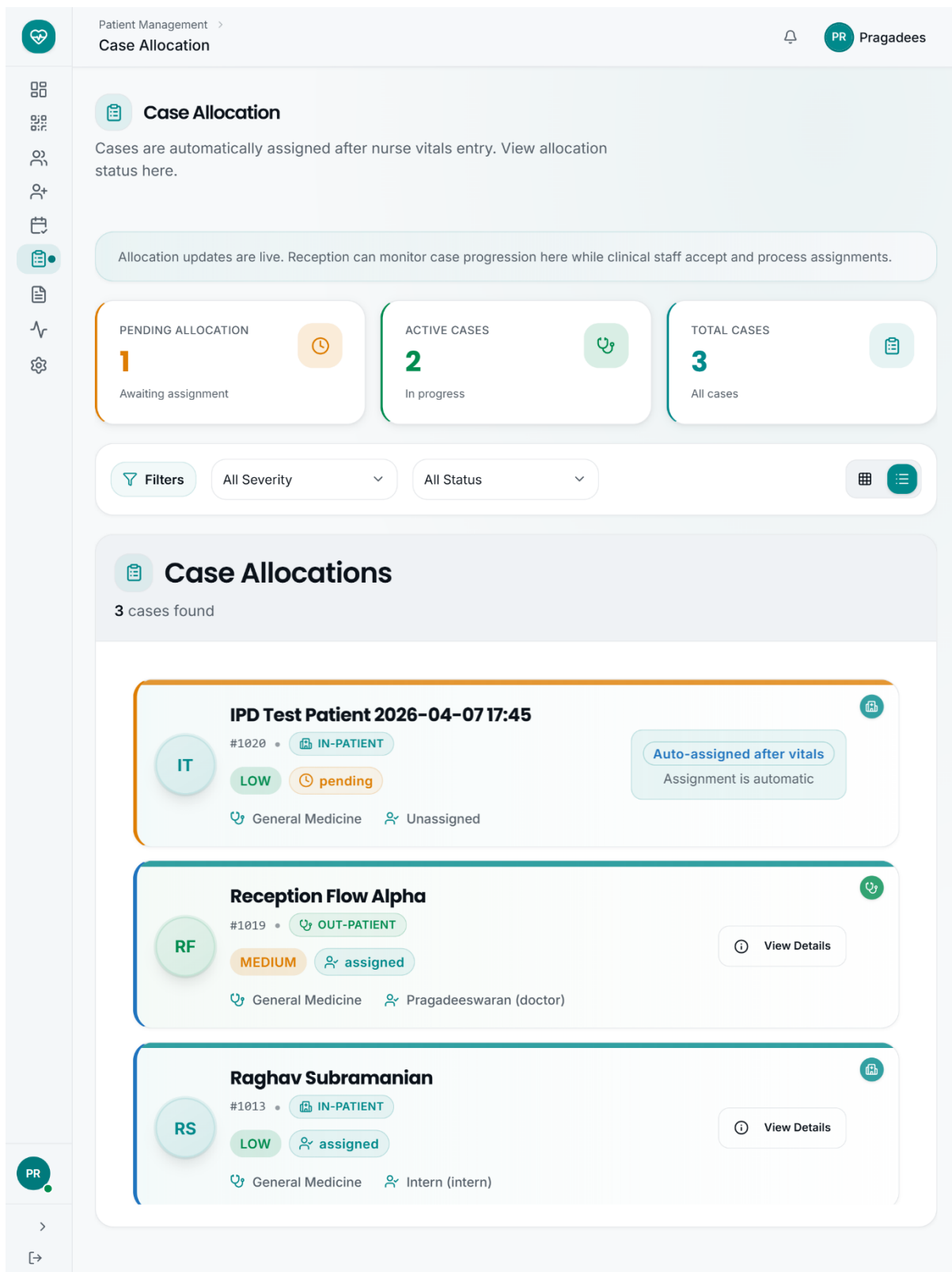


Figure 2.42 UI Design for Case Allocations Page

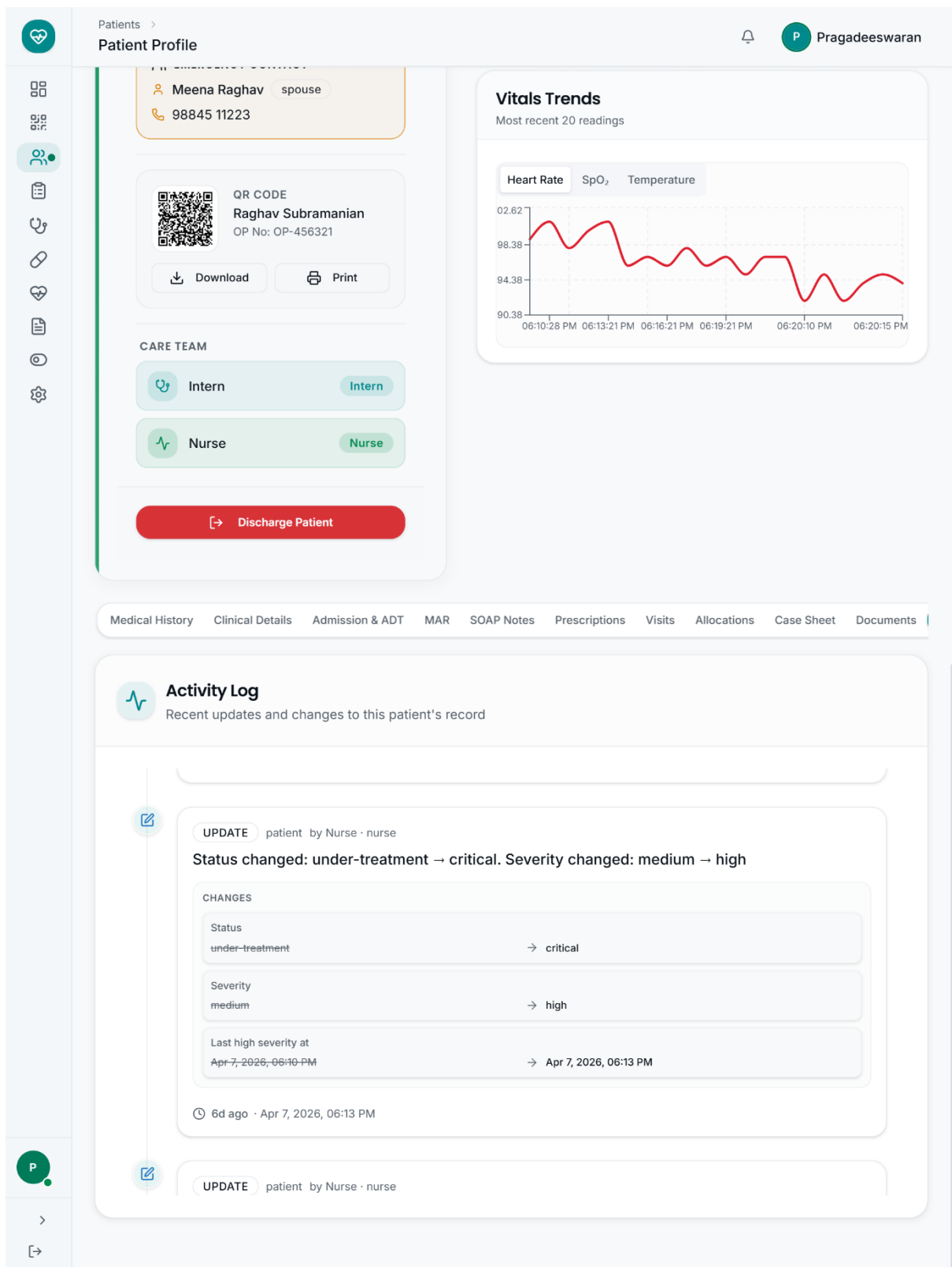


Figure 2.43 UI Design for Activity Log Page

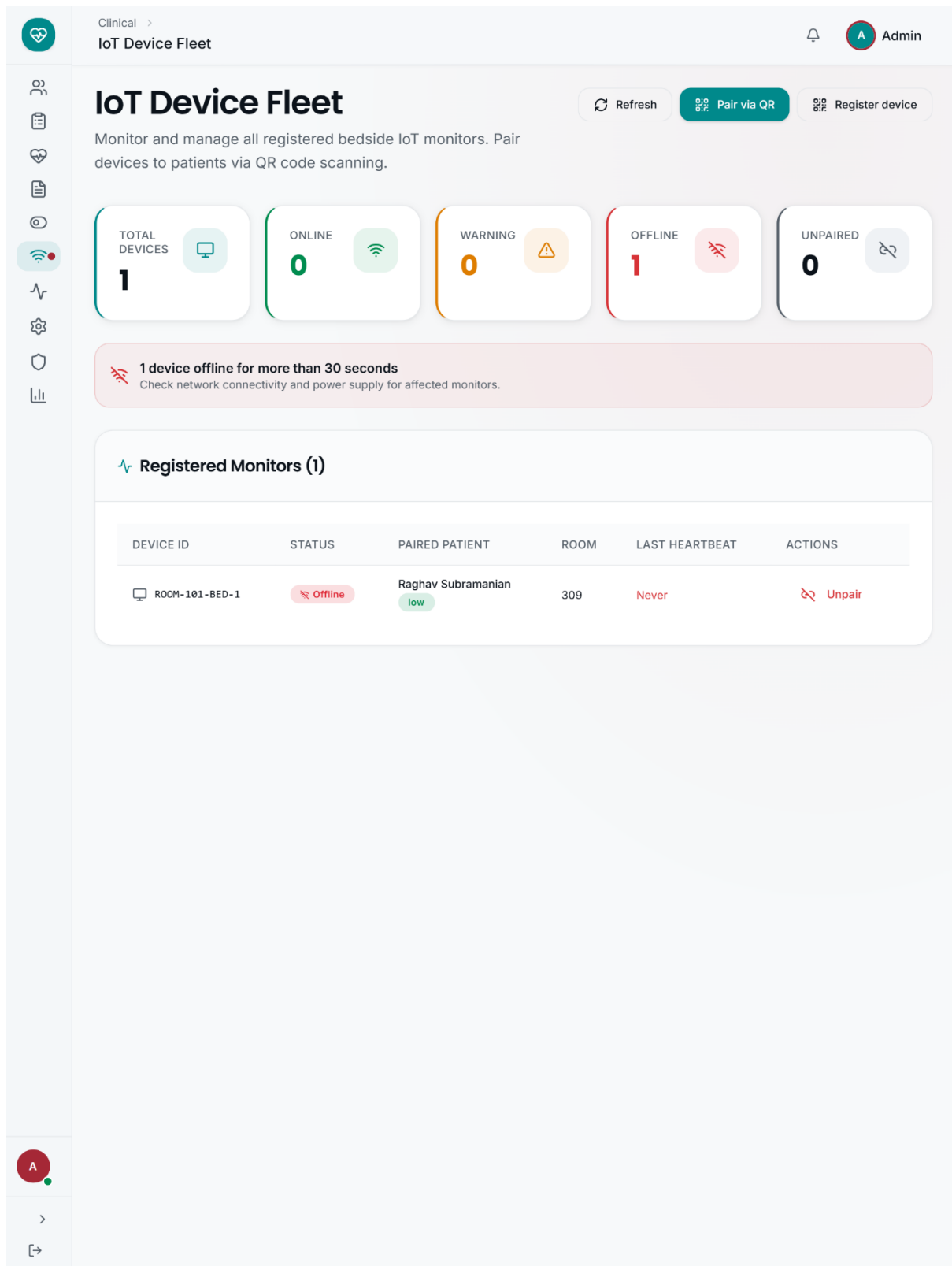


Figure 2.44 UI Design for IoT Device Fleet Page

2.3.5 Functional Test Cases

Table 2.13 Detailed Functional Test Case for Sprint 3

Feature	Test Case	Steps to Execute Test Case	Expected Output	Actual Output	Status	More Information
Severity from vitals	Thresholds, display, fallback	Open vitals; drive values across thresholds; observe severity and alerts; exercise fallback when the server severity call fails (test configuration).	Severity and alerts match rules; no crash on fallback; logs without PHI; diagnostics safe.	As expected,	Passed	Intern access rules from earlier sprints still apply.
Case allocation	Auto-assign, manual route, concurrency	Save vitals with auto-allocation enabled for High severity; use manual case allocation; repeat submit quickly or from a second session.	Non-intern assignee or clear error for High where rules require; duplicate attempts return a defined message; list matches storage after refresh.	As expected,	Passed	Manual path at /case-allocation when implemented.
Audit & activity	Immutable rows; live list	Perform a mutating action; open activity log; inspect newest row; keep activity open and trigger a remote auditable action.	Actor, time, entity, snapshot when built; new row without full reload when Realtime is on.	As expected,	Passed	Filters used for the test are recorded.
Dashboards, admin, IoT, vitals	Role landings and route gates	Open role dashboard; open /admin and /admin/analytics as administrator; doctor attempts same URLs; open /vitals and /iot-fleet for nurse vs doctor as per matrix.	Intake vs admin tiles: doctor blocked from admin KPI routes; vitals and fleet load or show empty state; no endless spinner.	As expected,	Passed	Exact labels may vary by branch.
Patient & visit status	Transitions and visibility	As doctor, intern, or administrator, apply allowed status changes; refresh boards/lists; optional concurrent update attempt.	Only permitted transitions; status visible on dashboards/lists; material changes leave audit evidence.	As expected,	Passed	States follow product policy.

2.3.6 Daily Call Progress

Table 2.14 Standup meetings for Sprint 3

Date	Day	Sprint Phase	Tasks Discussed / Completed
Mar 6, 2026	Fri	Sprint 3 start	Activity log merged timeline; audit filters; live list updates.
Mar 12, 2026	Thu	Severity & dashboards	Severity classification path; role dashboard content; admin landing.
Mar 18, 2026	Wed	Allocation	Provider selection rules; workload; intern exclusion on High; override flows.
Mar 25, 2026	Wed	Mid-sprint integration	Allocation audit logging; concurrent allocation QA; dashboard performance.
Apr 5, 2026	Sun	Audit & analytics	Audit taxonomy; KPI widgets; indexed queries for admin analytics.
Apr 12, 2026	Sun	Fleet & vitals	IoT fleet pages; vitals monitor consolidation; empty-state copy.
Apr 20, 2026	Mon	Hardening	Lint and tests; security checklist; accessibility on primary modals.
Apr 25, 2026	Sat	Regression	End-to-end rehearsal; README golden path; demo environment freeze.
Apr 30, 2026	Thu	Sprint close	Final fixes; stakeholder demo; retrospective; documentation update.

2.3.7 Committed Vs Completed User Stories

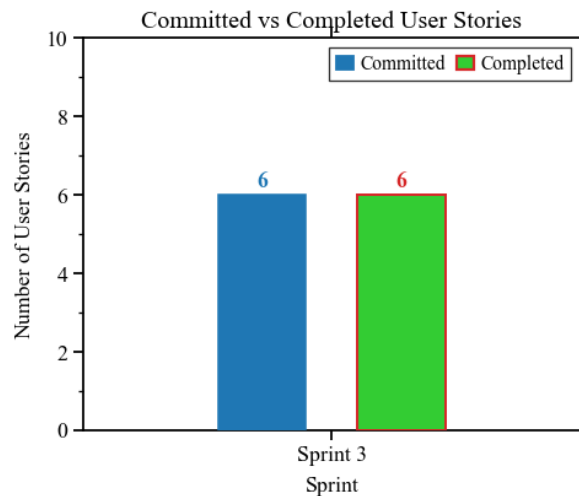


Figure 2.45 Bar graph for Committed Vs Completed User Stories for Sprint 3

2.3.8 Sprint Retrospective

Table 2.15 Sprint Retrospective for the Sprint 3

Liked	Learned	Lacked	Longed For
Severity-from-vitals and allocation closed with clear intern versus doctor behaviour on high severity.	Concurrent allocation needs deliberate UI debounce and idempotent server behaviour.	Double-submit edge cases cost extra QA time.	Stronger integration tests for rapid repeat clicks on allocation.
Audit plus activity in one pane helped reception and admin; realtime inserts felt responsive.	Long merged lists need sensible default date windows and reset controls.	Scroll fatigue when filters mis-set.	Saved filter presets and default last-24-hours window.
Role dashboards and admin analytics reduced duplicate landings; vitals dashboard consolidated monitoring.	Cold-load cost of aggregate KPI queries on large sets.	First-load slowness on analytics without warm-up.	Explain plans, indexes, or rollups before data volume grows.
Hardening caught dependency drift early; README path matched demo script.	Chart tooltips still weak for keyboard users on secondary views.	Accessibility gaps on non-modal charts.	Accessible table toggle or focus fixes for vitals and alert views.

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Project Outcomes

SerenDoc was delivered as a role-based care coordination web application built on a React client and Supabase-backed services. Across three sprints, the product moved from secure onboarding and patient–visit–document foundations to clinical workflows with append-only documentation, severity-aware access, and vitals-aware surfaces, and finally to governance features including vitals-informed severity classification with fallback behaviour, case allocation, tamper-evident audit and operational activity views, role and administrator dashboards, IoT fleet oversight where implemented, patient and visit status management, and release hardening for demonstration and review.

By the end of Sprint 3, authorised staff could follow a coherent path from registration and visit intake through clinical documentation, allocation decisions, monitoring views, and audit or activity review—without exposing full functionality to every role. The user interface was shaped per role (reception, nursing, doctor, intern, administrator), with realtime updates where configured for vitals and activity, OCR-assisted capture at registration with human review, and controlled sharing surfaces for case sheets where in scope. Feedback and operational visibility were supported through activity and audit mechanisms rather than through a separate maintenance-specific “session” metaphor, aligning with hospital workflows.

Key deliverables

- End-to-end patient and visit lifecycle support: authentication, registration, documents, clinical notes, prescriptions, case sheets, QR identification, vitals, allocation, dashboards, audit and activity, and status transitions—enforced with row-level security and role policies.
- Interpretable, auditable workflows: append-only clinical history, governance-oriented audit logging, and operational activity views for appropriate roles.
- Demonstrable quality path: linting, tests, security checks, accessibility attention on primary flows, and a documented rehearsal path suitable for stakeholder demonstration without real patient identifiers in test data.

In practice, the build keeps care coordination consistent across roles, holds sensitive data to policy-backed rules, and leaves hooks for more devices or interoperability later if a site needs them.

3.2 Hardware Implementation

The hardware component of the proposed system enables real-time patient monitoring and integration with the SerenDoc platform. It consists of a microcontroller connected to sensors that measure physiological parameters such as temperature and heart rate. These sensors continuously collect data and send it to the microcontroller for initial processing. The microcontroller processes the data and transmits it to the backend system using communication technologies like Wi-Fi or Bluetooth. The data is then stored and analysed in the system, where it is displayed through dashboards for monitoring. This integration ensures seamless communication between the hardware and software, allowing healthcare professionals to track patient conditions in real time. Overall, the hardware module enhances monitoring efficiency and supports timely clinical decision-making.

3.3 Committed Vs Completed User stories

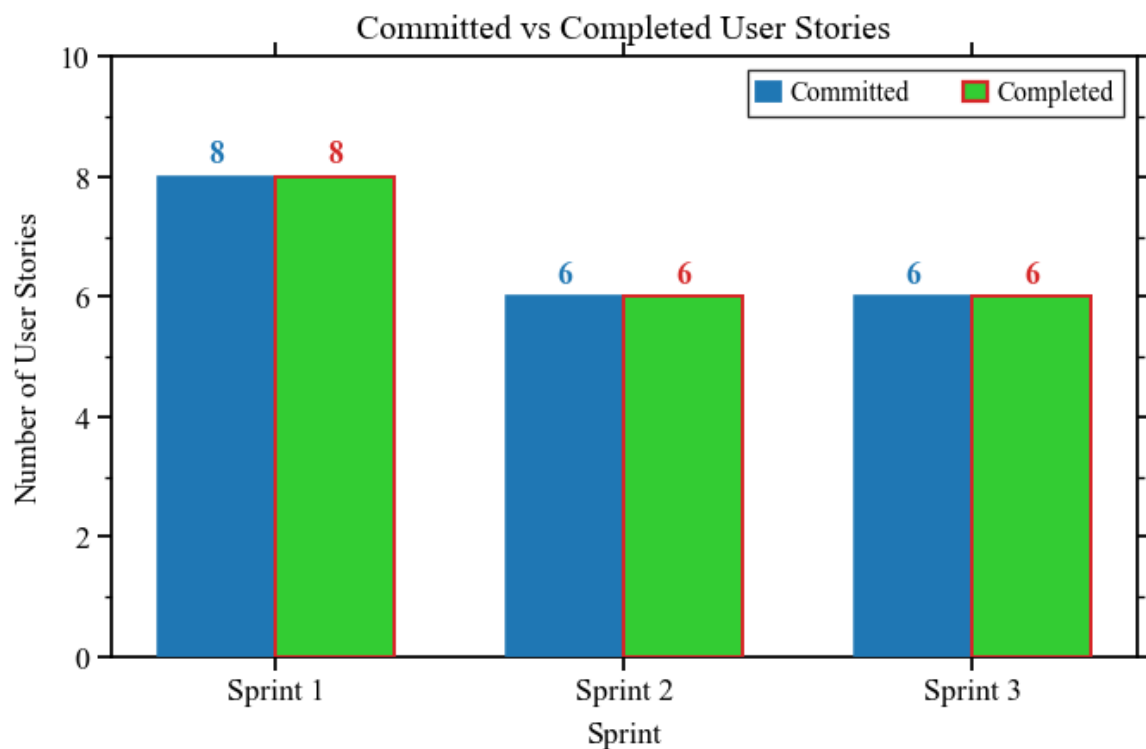


Figure 3.1 Bar graph for Committed Vs Completed User Stories for Sprint 1,2,3

CHAPTER 4

CONCLUSION & FUTURE ENHANCEMENTS

SerenDoc demonstrates coordinated, role-based care documentation well in development, but live deployment raises issues of connectivity, device limits, load, and changing clinical policy. The improvements below are practical next steps.

Ward conditions, connectivity, and devices

Realtime vitals, activity feeds, and uploads assume stable sessions; wards often have weak Wi-Fi, shared PCs, and interrupted work. Future work should add safe degraded modes (e.g. cached read-only views and clear resync after reconnecting), tablet-friendly layouts for bedside tasks, and explicit success or retry messaging when OCR or saves fail so staff know what reached the record.

Client footprint, realtime traffic, and backend scale

Heavy dashboards and Realtime subscriptions can slow older ward machines when update volume is high. Improvements include lazy loading of large routes, careful use of live channels, indexed queries for allocation and analytics, optional KPI rollups for administrators, and bounded Edge Functions so spikes in vitals or audit traffic do not block urgent registration or clinical writes.

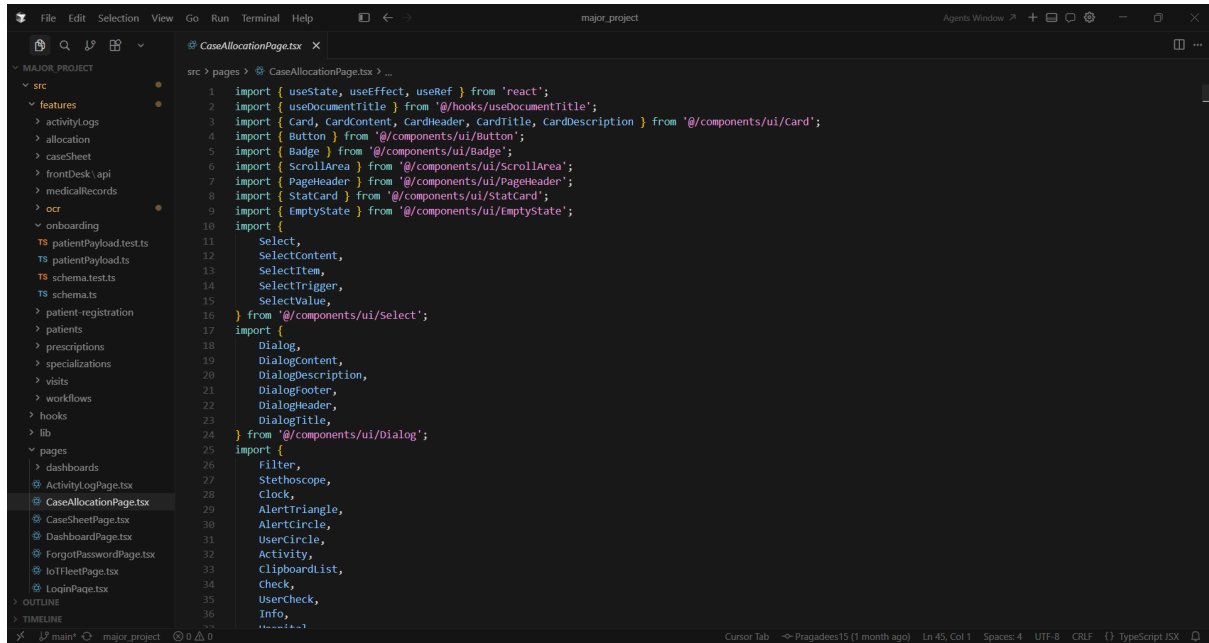
Maintainability of rule-based access and severity logic

Row-level rules, severity bands, and allocation logic stay manageable when policy is small and stable; they become harder to test and explain as roles and pathways change. Versioned policy, pre-rollout simulation, and—where governance allows—limited data-assisted signals with human review can help, without replacing local clinical authority.

Together, these directions stress reliability and day-to-day fit while keeping access control, traceability, and role-based coordination intact.

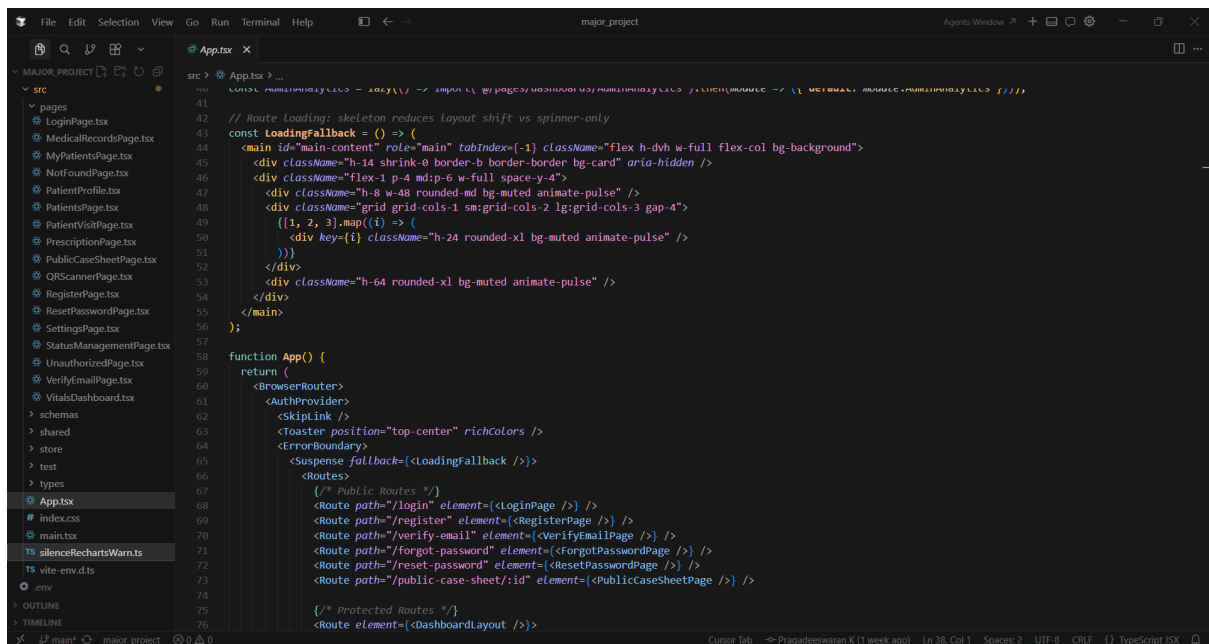
APPENDIX A

SAMPLE CODING



```
1 import { useState, useEffect, useRef } from 'react';
2 import { useDocumentTitle } from '@hooks/useDocumentTitle';
3 import { Card, CardContent, CardHeader, CardTitle, CardDescription } from '@components/ui/Card';
4 import { Button } from '@components/ui/Button';
5 import { Badge } from '@components/ui/Badge';
6 import { ScrollArea } from '@components/ui/ScrollArea';
7 import { PageHeader } from '@components/ui/PageHeader';
8 import { StatCard } from '@components/ui/StatCard';
9 import { EmptyState } from '@components/ui/EmptyState';
10 import {
11   Select,
12   SelectContent,
13   SelectItem,
14   SelectTrigger,
15   SelectValue,
16 } from '@components/ui/Select';
17 import {
18   Dialog,
19   DialogContent,
20   DialogDescription,
21   DialogFooter,
22   DialogHeader,
23   DialogTitle,
24 } from '@components/ui/Dialog';
25 import {
26   Filter,
27   Stethoscope,
28   Clock,
29   AlertTriangle,
30   AlertCircle,
31   UserCircle,
32   Activity,
33   ClipboardList,
34   Check,
35   UserCheck,
36   Info,
37 } from '@components/ui/Icons';
```

Figure 4.1 Code screenshot 1



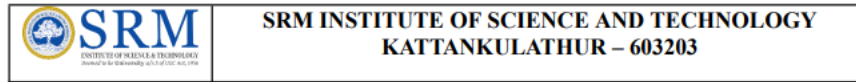
```
41 // Route loading: skeleton reduces layout shift vs spinner-only
42 const LoadingFallback = () => {
43   <main id="main-content" role="main" tabIndex=-1 className="flex h-dvh w-full flex-col bg-background">
44     <div className="h-14 shrink-0 border-b border-border bg-card aria-hidden />
45     <div className="flex-1 p-4 md:p-6 w-full space-y-4">
46       <div className="h-8 w-48 rounded-md bg-muted animate-pulse" />
47       <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-4">
48         {[1, 2, 3].map((i) => {
49           <div key={i} className="h-24 rounded-xl bg-muted animate-pulse" />
50         })}
51       </div>
52       <div className="h-64 rounded-xl bg-muted animate-pulse" />
53     </div>
54   </main>
55 );
56
57 function App() {
58   return (
59     <BrowserRouter>
60       <AuthProvider>
61         <SkiptLink />
62         <Toaster position="top-center" richColors />
63         <ErrorBoundary>
64           <Suspense fallback={<LoadingFallback />}>
65             <Routes>
66               </* Public Routes */>
67               <Route path="/login" element={<LoginPage />} />
68               <Route path="/register" element={<RegisterPage />} />
69               <Route path="/verify-email" element={<VerifyEmailPage />} />
70               <Route path="/forgot-password" element={<ForgotPasswordPage />} />
71               <Route path="/reset-password" element={<ResetPasswordPage />} />
72               <Route path="/public-case-sheet/:id" element={<PublicCaseSheetPage />} />
73             </Routes>
74             </* Protected Routes */>
75             <Route element={<DashboardLayout />} />
76           </Suspense>
66         </ErrorBoundary>
67       </AuthProvider>
68     </BrowserRouter>
69   );
70 }
```

Figure 4.2 Code screenshot 2

Figure 4.1 and 4.2 Shows the sample coding for the project

APPENDIX B

PATENT APPLICATION



Invention Disclosure Form

To be filled by the inventors

Please provide highly relevant information for details asked below and use consistent language while describing the specific feature or element in the invention disclosure.

1. **Title of invention** (Please indicate a title for the invention and technology of the invention)

SerenDoc: A Human-Centric AI and IoT-Based Clinical Workload Orchestration System for Smart Healthcare

2. **Describe the invention.** (Please describe specifically about the general purpose of invention. Is the invention a new process, device or product, system, software or a combination of these elements?)

The general purpose of the invention is to transform manual and fragmented clinical workflows into an intelligent, unified healthcare operation by automating medical documentation, enabling continuous patient monitoring, and optimising clinician workload in real time. The invention is a combination of a system, software, and process, supported by an IoT-based device layer. It introduces a coordinated workflow in which handwritten medical case sheets are captured and processed through AI-driven OCR and natural language processing to generate structured digital clinical records. In parallel, IoT-enabled sensing devices continuously acquire patient physiological parameters and transmit them to the system for real-time analysis. The invention further incorporates an AI-based severity assessment and fatigue-aware workload orchestration process that dynamically assigns and balances clinical tasks among healthcare professionals based on patient risk, workload distribution, and shift continuity. The system is implemented through a full-stack software architecture consisting of frontend dashboards, backend services, and modular machine-learning components, while the IoT layer functions as an interoperable hardware interface for patient data acquisition. Collectively, the invention operates as an integrated smart healthcare ecosystem rather than a standalone device or application, with the objective of improving clinical efficiency, decision-making timeliness, and clinician well-being.

SRMIST - Confidential

2

Figure 4.3 Patent Invention Disclosure Form for Smart Healthcare System

APPENDIX C

PAPER PUBLICATION

SerenDoc: A Human-Centric AI and IoT-Based Clinical Workload Orchestration System for Smart Healthcare

Dr. Vinston Raja R Assistant Professor <i>Dept. of Computational Intelligence</i> <i>SRM Institute of Science and Technology</i> Kattankulathur, Chennai, India vinstonr@srmist.edu.in	Mr. Shyam J Student <i>Dept. of Computational Intelligence</i> <i>SRM Institute of Science and Technology</i> Kattankulathur, Chennai, India jshyam2005@gmail.com	Ms. Aditi V Student <i>Dept. of Computational Intelligence</i> <i>SRM Institute of Science and Technology</i> Kattankulathur, Chennai, India aditivasudevan25@gmail.com
Mr. Pragadeeswaran K Student <i>Dept. of Computational Intelligence</i> <i>SRM Institute of Science and Technology</i> Kattankulathur, Chennai, India pragadees1323@gmail.com		

Abstract—Healthcare delivery systems, especially in the less developed areas, are still very much dependent on handwritten medical records and manually driven clinical workflows. These kinds of practices result in fragmented patient information, increased documentation burden, delayed clinical decision-making, and significant physician burnout. Though current Electronic Medical Record (EMR) platforms are aimed at data digitisation, they do not, in most cases, integrate physiological monitoring in real-time and intelligent workload management; thus, they have a limited impact on operations. This article introduces SerenDoc, a human-centred smart healthcare ecosystem that combines AI (Artificial Intelligence), IoT (Internet of Things), OCR (Optical Character Recognition), and NLP (Natural Language Processing) technologies to automate clinical documentation, monitor patient vitals continuously, and manage clinician workloads intelligently. SerenDoc converts the images of handwritten case sheets into digital text through context-aware OCR and medical NLP pipelines, whereas IoT-enabled sensors monitor the patient vitals like heart rate, oxygen saturation, and body temperature in real-time. Moreover, the new AI-based severity classification and fatigue-aware task allocation engine can reassign clinicians with new cases that are based on the patient's risk, the workload balance, and the shift continuity. The SerenDoc system utilises a scalable full-stack architecture consisting of a React-based frontend, a Supabase-powered backend, and lightweight machine learning services. The system was verified in a simulated hospital setting and reached Technology Readiness Level (TRL) 5-6. The performance evaluation with mock datasets shows the high OCR accuracy (greater than 90%), the efficient severity classification (F1-score greater than 0.85), and the lowered documentation time. SerenDoc provides a feasible way to the next generation of smart, clinician-aware healthcare systems that lead to the improvement of the efficiency, transparency, and the doctor's well-being.

Index Terms—Smart Healthcare, Clinical Workload Management, Optical Character Recognition, Natural Language Processing, Internet of Things, Artificial Intelligence, Doctor Well-being, Digital Health

I. INTRODUCTION

Worldwide healthcare systems are facing extreme challenges to manage the ever-increasing number of patients, the shortage of medical personnel, and the higher demands for quality care. Even though medical-technological advancements have been made, a large part of hospital routines—especially documentation and patient monitoring - is still done manually and in an inefficient way. In numerous hospitals, particularly in developing countries, handwritten case sheets are still the most common method of recording patient history, diagnoses, and treatment plans. These handwritten records are most of the time unstructured, inconsistent, and far from being linked to the real-time patient vitals, which makes them very difficult to analyse, access, and share between different departments.

Handwritten documentation consumes a large portion of the clinicians' cognitive capacity and time. Doctors and interns are heavily involved in note-taking, updating case sheets, and doing paperwork, in which they spend a large portion of their working hours; thus, without realising it, they are indirectly deprived of patient care. This kind of administrative load is one of the main reasons for physician exhaustion and burnout, and, in turn, it leads to a decrease in diagnostic accuracy, decision-making quality, and patient safety. Plenty of research works have shown that overburdening physicians with documentation tasks leads to an increase in medical errors and a decrease in healthcare professionals' job satisfaction.

Electronic Medical Record (EMR) systems were basically designed to fix these problems by digitising patient data and making information more accessible. Nevertheless, the major part of current EMR schemes are primarily aimed at the mere entering of data and observing administrative requirements

Figure 4.4 First Page of the SerenDoc Research Paper

APPENDIX D

HARDWARE DESIGN AND IMPLEMENTATION

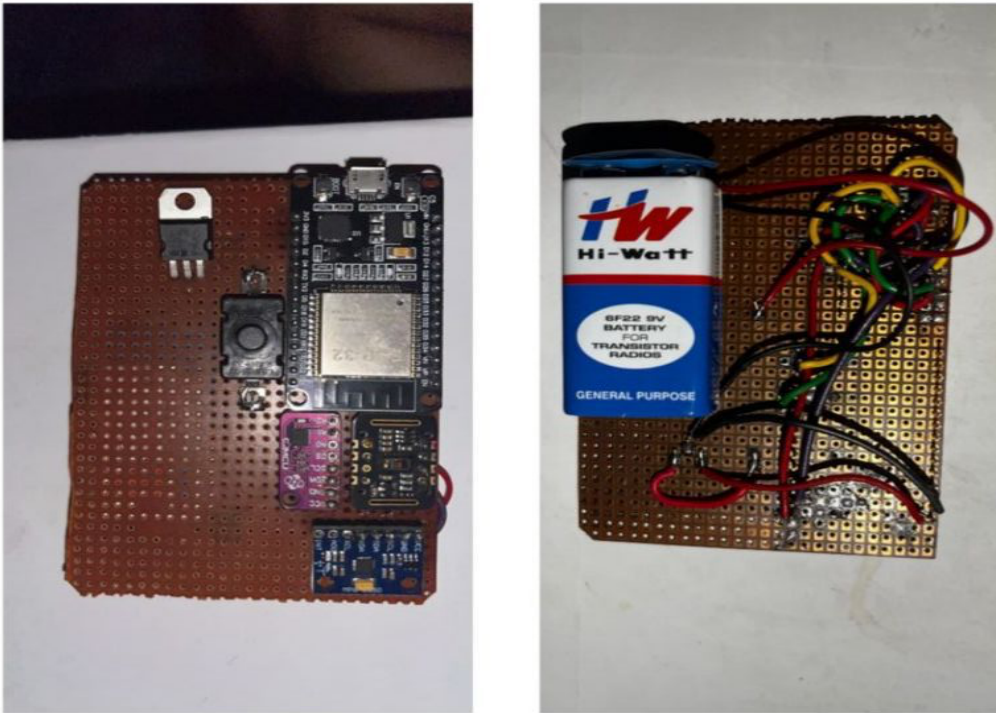


Figure 4.5 Prototype Multi-Sensor Module for Real-Time Patient Physiological Data Acquisition

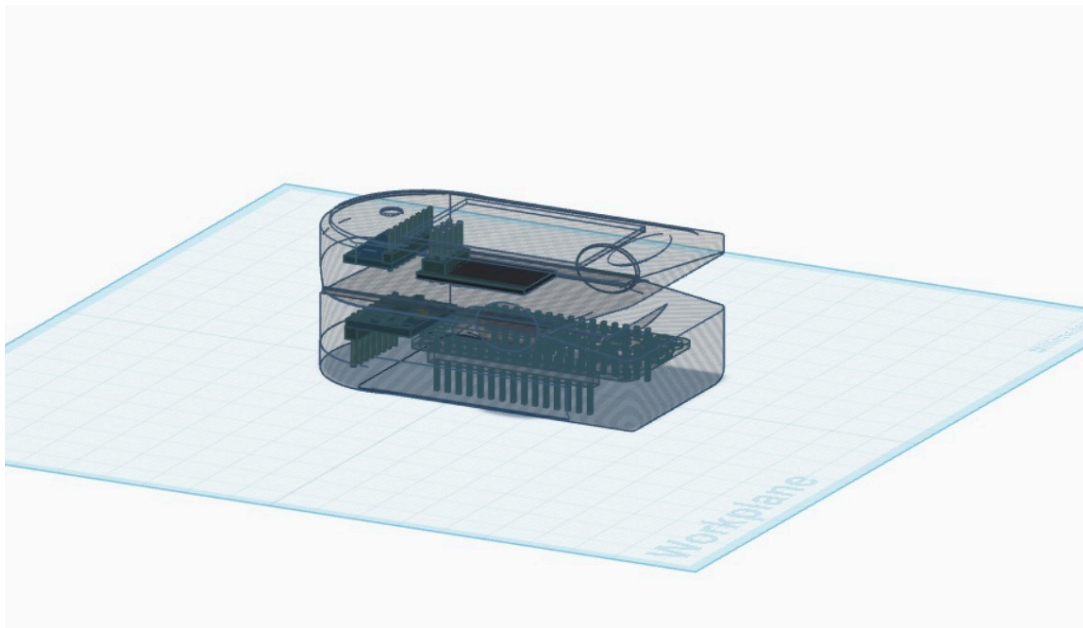


Figure 4.6 Three-Dimensional Internal Layout of the Proposed Wearable Multi-Sensor Patient Monitoring Device

APPENDIX E

PLAGIARISM REPORT

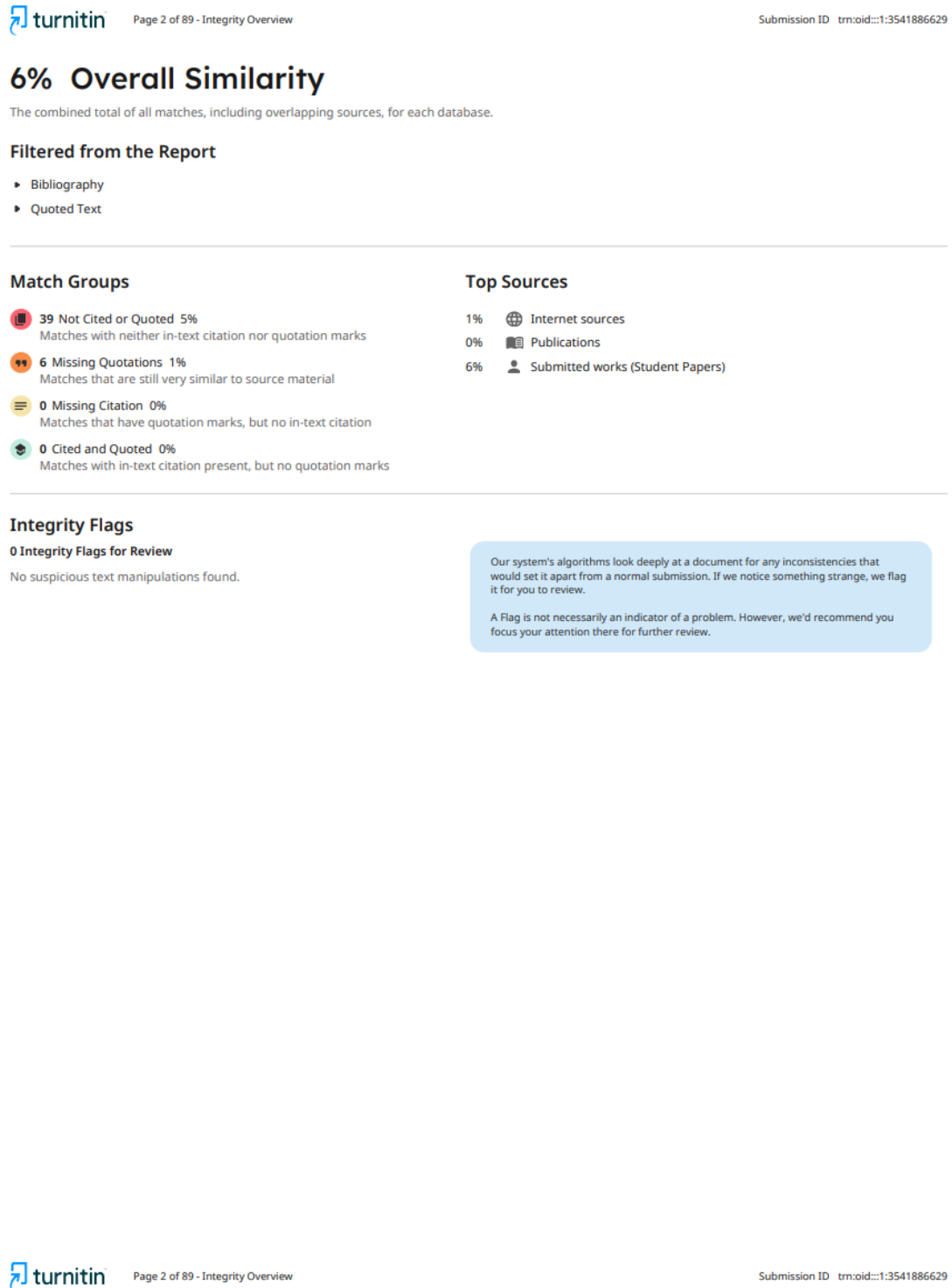


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